



Department of Education

STEM Education
National Schools of Excellence
Chemistry Students' Resource Book
Grade 12
Term 2

DOMAIN **Phylum**

SCIENCE **Chemistry**

Technology

Engineering

Mathematics

Issued free to schools by the Department of Education

Published in 2021 by the Department of Education, Papua New Guinea

© Copyright 2021, Department of Education, Papua New Guinea

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted by any form or by any means electronic, mechanical, photocopying, recording or otherwise without the prior written permission of the publisher.

ISBN

Acknowledgements

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

This document was developed with support from the General Education Services Division of the Department of Education.

Table of Contents

THEMA: ENERGY	1
MODULE 12.1: GREEN HYDROGEN ENERGY ECONOMY	3
Topic 1: Introduction to Hydrogen Energy Economy	6
Topic 2: Methods of Producing Hydrogen	12
Topic 3: Storing Hydrogen	27
Topic 4: Hydrogen Fuel-Cell.....	37
Sample Module Activities	44
THEMA: BATTERY TECHNOLOGY	48
MODULE 12.1: SOLID-STATE BATTERY.....	50
Topic 1: Introduction to Solid-State Battery	51
Topic 2: Solid Electrolyte.....	57
Topic 3: Manufacturing of Solid Batteries	60
Sample Topic Activities	78
References	80

THEMA: ENERGY

Introduction

Energy is the ability to do work. It comes in several forms. It may exist in potential, kinetic, thermal, electrical, chemical, nuclear, or other forms of energy. However, the focus will be mainly on chemical energy in chemical reactions. In chemical bonds, the energy is stored in the bonds of chemical compounds. During a chemical reaction, chemical energy may be taken in or given off often in the form of heat.

Reactions that give off energy during a chemical reaction are said to be exothermic reactions while the ones that take in energy from the surrounding in order for the reaction to proceed are endothermic reactions. A good advantage of energy is that it can be transferred from a state to another without being destroyed. Some conversion of energy include the transfer of energy in food that we take in which is converted to mechanical energy and heat by the body, coal to electrical energy to supply electricity and conversion of chemical energy in batteries to supply energy through the process of electrolysis.

We can all agree that the sooner we move away from fossil fuels and toward other energy sources, the better. Two of the main reasons to move away includes Environmental pollution caused by non-renewable energy as well as the inevitability of the exhaustion of non-renewable energy sources. Therefore, we must look for ways to utilize the potential of renewable energy.

Listed are few of the many career pathways in the field of the energy industry:

- Civil Engineer
- Solar Photovoltaic Installer
- Petroleum Engineer
- Environmental Scientist
- Financial Analyst
- Electrical Power Line Installer
- Electrical Engineer
- Geoscientist
- Oil and Gas Service Unit operators
- Wind Turbine Technician
- Power Plant Operator; etc.,

MODULE 12.1: GREEN HYDROGEN ENERGY ECONOMY

4-5 weeks

Currently, the most hailed renewable energy technology that has been considered to be the future of energy is the **Hydrogen Fuel/Energy Technology**. Hydrogen is an energy carrier, and due to this property, it can be used as fuel that can be converted to any form of energy, including electricity or heat. Ultimately, it has the potential to meet power-demands of an economy and solve some of humanity's energy problems.

Presently, humanity is predominantly supported by the hydrocarbon-based economy, meaning most of our power demands are being met by burning non-renewable resources such as fossil fuels. However, as we have learnt, this type of energy economy has led to environmental pollution. Many countries, upon learning about the potential of hydrogen energy technology, are making plans to transition from hydrocarbon-based energy economy to a hydrogen-based energy economy. A **Hydrogen-Based Energy Economy** is a futuristic economy whose energy delivery system is based on hydrogen. This type of economy promises a more carbon neutral and net-zero pollution results. If integrated with other renewable energy sources, this could result a **Green Hydrogen Energy Economy** where humanity's power demands will be met predominantly by renewable energy sources. Currently, research is being done on how we can maximize the efficiency of renewable energy systems so that humanity can solely depend on renewable energy.

Why do we need to study this?

As mentioned, hydrogen energy technology is the future of energy and it has the potential to solve some of humanity's global energy problems. Papua New Guinea (PNG) has the potential to contribute to solving both these problems on a national and global scale. We are rich with numerous energy resources, both non-renewable and renewable. If we adopt this hydrogen energy technology, it will only diversify our energy options.

Here is a story to help you understand how Papua New Guinea has the potential to use Hydrogen Energy as our fuel.

Andrew Forrest, the iron ore billionaire, has announced plans to establish a vast green energy manufacturing complex in northern Queensland, initially focused on hydrogen electrolyzers before expanding to wind turbine equipment, solar PV cells, and electrical wiring. The announcement, made by Fortescue Future Industries and the Queensland State Government, ushers Australia into a new era of green industry manufacturing and energy production centers, and its location in Gladstone, the state's coal and LNG hub, is no coincidence. Hundreds of jobs will be produced during construction and over the project's lifetime, but thousands more could be created when the center grows into the manufacturing of wind turbines, solar PV cells, and other infrastructure, assisting Australia's transformation through green industries. (Parkinson, Giles, 2021)

"As we transition away from fossil fuels, FFI's aim is to become the world's top integrated, 100 percent renewable energy and green products organization, powering the Australian economy and creating jobs," said Julie Shuttleworth, CEO of Fortescue Future Industries. Their manufacturing arm will start with hydrogen electrolyzers and expand into additional green energy industry products in the future.

The government of Papua New Guinea has signed a historic agreement with one of the world's largest firms to collaborate on "Green Energy," a partnership that is expected to transform power generation and distribution in the near future.

The government today signed a "Deed of Agreement" with Australian corporation Fortescue Metals Group (FMG), the world's fourth largest producer of iron ore, in yet another bold attempt to "Take Back PNG."

Minister of State Enterprises Hon. William Duma and FMG Founder/Owner Dr. Andrew Forrest signed the instrument, which was witnessed and co-signed by Prime Minister Hon. James Marape and Fortescue Future Industries CEO Julie Suttleworth.

The agreement, according to Prime Minister Marape, is "historic" since it allows FMG's industry company Fortescue Future Industries (FFI) and PNGFFI to study and eventually look into developing renewable energy sources like as hydro, solar, wind, and geothermal power.

As the globe moves toward green energy and attempts to reduce its carbon footprint, he said this was the "most suitable and relevant route to proceed." PM Marape further stated that his Pangu-led government is committed to developing renewable energy from the country's natural resources, which include solar, wind, gas, and geothermal hot springs.

As it continues to explore various sites in PNG, including geothermal, hydro, and solar, the investor is eyeing locations in Gulf and West New Britain provinces to research and evaluate hydropower and geothermal potentials.

PM Marape claimed that during one of their earlier meetings, Dr. Forrest indicated that his company is capable of producing over 25 gigawatts (250 billion watts) of sustainable "green hydrogen and green ammonia" energy from PNG, which is sufficient for both internal usage and export to global markets. (PNG Business News, 2021)

As a result, Papua New Guinea is assisting other countries in making the transition to a Green Economy. If Papua New Guinea's hydrogen fuel can help Australia shift to a green economy, why couldn't we use it to help us adapt and meet our energy needs as well? We have a great deal of promise. Renewable energy sources have taken center stage as PNG and other countries aim to meet 32 percent of their national power demand with renewable energy sources by 2030. In addition, PNG has set a goal of providing electricity to 70% of the country's dispersed population by 2030.

Now that we know we have the ability to develop a hydrogen economy, let's get started. Let's get started learning about Hydrogen Energy Technology and the Hydrogen Economy.

The key elements of the Hydrogen Economy are:

- Production of Hydrogen
- Packaging of Hydrogen
- Transportation of Hydrogen
- Storage of Hydrogen

- Transfer of Hydrogen

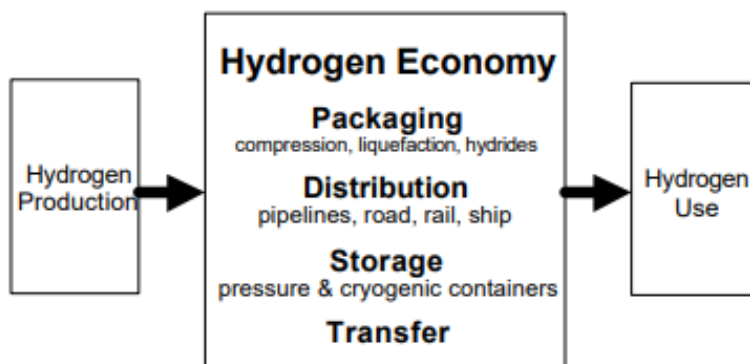


Figure SC-E-SRB-1: Elements of Hydrogen Economy

The topics of this module will be focused around the listed key elements above.

Topic 1: Introduction to Hydrogen Energy Economy

Key Points

- Hydrogen is an energy carrier; it has the potential to power an economy.
- Hydrogen is found in H₂ carriers (molecules or compounds that contain high-energy-density hydrogen).

1.1 Hydrogen as Fuel

Hydrogen is the first element on the periodic table. Its atom contains 1 proton and 1 electron. In its gaseous state (H₂), it is highly combustible, especially when it reacts with oxygen gas (O₂), however, it is not poisonous.

Hydrogen is the most simple and abundant element one can find in the universe. However, you will never find it alone in its purest form (H₂ or H). Naturally, hydrogen is found in combination with other elements, such as water (H₂O) where it is combined with Oxygen, hydrocarbons such as oil and fossil fuels, carbohydrates, proteins, hybrids, and other molecules. One reason why hydrogen is rarely found alone is because it is readily reactive so it tends to combine with other elements very easily. Another reason is because it is a very light element (15 times lighter than air), it tends to escape towards space where it is dispersed.

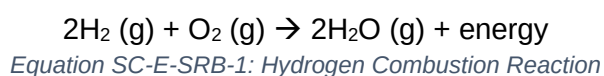
Majority of hydrogen is found in molecular forms everywhere on Earth in different forms – solid, liquid and gaseous. You can find it in land, air and in our large oceans and seas. It is even found in space. For example, just within our Solar System, 91.0% of our Sun is made up of hydrogen in its plasma state and Jupiter and Saturn contain mostly hydrocarbon.^{1 2}

Hydrogen as Energy Carrier

Our desire is to work with Hydrogen in its purest form. This will require us to separate Hydrogen from other elements in order to obtain isolated Hydrogen. The separation of pure hydrogen necessitates a significant amount of energy. This is why we say hydrogen is not a primary source of energy like coal, petroleum, or methane, but rather an energy carrier like electricity, which is a kind of energy that does not occur naturally.

An energy carrier is a substance (fuel) or a phenomenon (energy system) that holds energy that may be transferred into other forms, such as mechanical work or heat, to power chemical or physical processes.

Let's look at the Hydrogen Combustion Reaction to see how Hydrogen functions as an energy carrier or "fuel." Pure H₂ (g) combines with pure O₂ (g) to generate water vapor (H₂O) and release energy in a flame of pure H₂ (g) fuel and pure O₂ (g).



¹ [Hydrogen - Wikipedia](#)

² [In Depth | Sun – NASA Solar System Exploration](#)

When employed in a combustion reaction, hydrogen is a pure fuel that produces only water, energy, and/or heat. This type of hydrogen combustion is commonly utilized to propel space shuttles into orbit.

NOTE: If this reaction occurs in atmospheric air rather than pure O₂, as is usually the case, tiny amounts of nitrogen oxides and water vapor may be produced.

Therefore, hydrogen is an energy carrier, meaning it has the ability to transport and/or store a vast quantity of energy.

H₂ Carriers

H₂ Carriers are a source of hydrogen or they contain the element Hydrogen in their chemical structure. H₂ Carriers basically refer to any chemical product or molecule with a high energy density that may be used to liberate hydrogen on demand. They can exist in liquid or solid states. Methanol, Ethanol, Oil, Coal, Natural Gas, Biogas, Biofuels, Ammonia, Carbohydrates, Hydrocarbons, Water, and so on are the most common high energy-density H₂ carriers that can provide energy to end-users. Henceforth, when looking into Hydrogen Energy technology, identify all potential H₂ carriers that can source your Hydrogen.

1.2 Introduction to Hydrogen Economy

1.2.1 Rationale

At the moment, the world is primarily trapped into a "Fossil Fuel Economy" or "Hydrocarbon-based Energy Economy." Our cars, trains, and planes are nearly entirely powered by petroleum products such as gasoline (petrol) and/or diesel. Oil, natural gas, and coal are used to fuel a large majority of the world's power plants. The majority of countries' economies are currently reliant on fossil fuels.

While fossil fuels have played a significant role in bringing society to where it is today, they also bring with them four major issues:

1. Air Pollution
2. Environmental Pollution
3. Global Warming
4. Dependence

Three of the four are linked to pollution in the environment as a whole. We'd want to concentrate on the fourth concern, which is reliance. Most countries can't generate enough oil to meet their demands, therefore they have to import it from oil-producing countries. This leads to a dependency on money. When Middle Eastern oil producers decide to raise their prices, the rest of the world is forced to pay the higher price.

As a result, transitioning one's economy from a hydrocarbon-based economy to a hydrogen-based economy is one of the alternative options to stop contributing to environmental degradation and a nation's economic dependence on fossil fuels.

1.2.2 Hydrogen-based Economy

The term "Hydrogen Economy" refers to a hydrogen-based (H_2) energy distribution system that could be implemented in the future. In 1970, during a conversation at the General Motors (GM) Technical Center, John Bockris coined the word. Proponents of the proposed technology promote hydrogen as a viable fuel source. Pure hydrogen, like other energy sources, does not occur in great quantities naturally, but it can be made using a variety of methods. As a result, hydrogen serves as an energy carrier rather than a primary source of energy. The viability of a hydrogen economy is influenced by the use of fossil fuels, the generation of sustainable energy, and energy sourcing.

1.2.3 Elements of a Hydrogen Economy

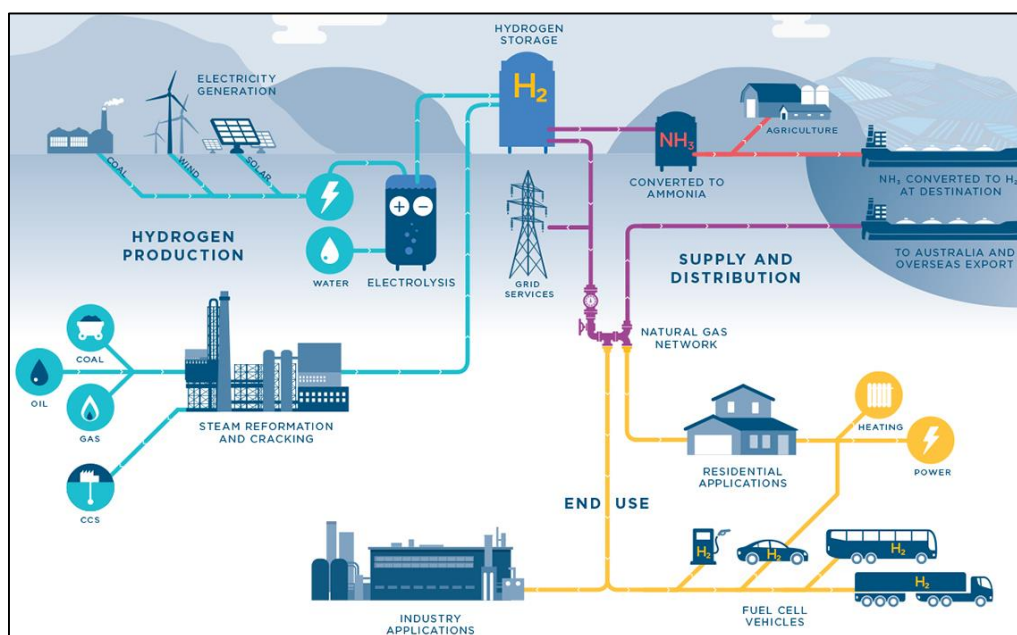


Figure SC-E-SRB- 1: A Sample Hydrogen Economy Strategy (Source Link:

https://wwwherbertsmithfreehills.com/sites/contenthub_mothership/files/hydrogen-world-infographic-v5_1170px.jpg)

The key elements of a hydrogen economy are:

- Production of Hydrogen
- Packaging of Hydrogen
- Transportation of Hydrogen
- Storage of Hydrogen
- Transfer of Hydrogen
- End-users

For a functional hydrogen economy, we need to take into consideration:

- How we are going to produce hydrogen?
- What are the methods to produces it?
- Will we use non-renewable or renewable resources as energy to drive the production of hydrogen?
- How do we safe store and package the hydrogen since it is flammable and highly reactive?
- How do we transport and transfer the hydrogen to the end-users?

- How do we integrate this into a energy deliver system that already exists and how do we make it more renewable and sustainable?

As we dive more into the topics of this module, we will discuss some of our options to help us identify the answer to these questions.

1.2.4 Comparison between Hydrogen Energy to Others

In the preceding section, we discussed the enormous global problems posed by the hydrocarbon-based economy. The hydrogen economy claims to be able to address all of the problems that the hydrocarbon economy creates. As a result, there are several advantages to a hydrogen-based economy:

- **Pollution generated by fossil fuels is eliminated** — when hydrogen is utilized in a fuel cell to generate electricity, it is a completely clean technology. Water is the only byproduct. Hydrogen also poses no threat to the environment.
- **No greenhouse gas emissions** - if hydrogen is produced by electrolysis of water, it emits no greenhouse gases into the atmosphere. A complete cycle exists: electrolysis produces hydrogen from water, and the hydrogen recombines with oxygen in a fuel cell to produce water and power.
- **Elimination of economic dependency** — Elimination of reliance on oil means no longer being reliant on the Middle East and its oil reserves.
- **Distributed production** - Hydrogen can be created wherever energy and water are available. It is also possible to make it at home using quite simple technology.

There is a strong push toward hydrogen since the problems with the fossil fuel economy are so severe, and the environmental benefits of the hydrogen economy are so significant. The hydrogen economy has the potential to change the current energy infrastructure by changing fuel demand from fossil fuels to hydrogen.

Technological Hurdles of Hydrogen Economy

The major question in the hydrogen economy is "Where does the hydrogen come from?" Following that comes the problem of transporting, distributing, and storing hydrogen. Hydrogen is big and difficult to manage in its natural gaseous state. Once both of these problems are solved in a cost-effective manner, the hydrogen economy will be in place.

In the next topic, the first issue, "Where does the hydrogen come from?" will be examined in relation to various methods of producing hydrogen.

1.3 Summary

- A functional hydrogen economy may include the following requirements: producing hydrogen, storing hydrogen, packaging hydrogen, and distributing hydrogen to end-users.
- Papua New Guinea has the potential to adopt a hydrogen energy economy to help diversify its energy resources. It may also help to solve PNG's and humanity's current and future energy demand challenges.

- Hydrogen is plentiful, yet it is only found in compounds or molecules, not in its purest form.
- The current hydrocarbon economy is becoming untenable due to rising demand and diminishing resources.
- Hydrogen is a fuel and/or an energy carrier. Because of its higher energy density and lower environmental impact, the hydrogen economy could serve as a replacement. However, the hydrogen economy is constrained by the difficulty of transporting and storing hydrogen. It's important to note that it's 'tough,' but not impossible. As future problem solvers and engineers engaged in the topic of energy, this is a task that we must take on and overcome for the sake of humanity.
- The risks associated with hydrogen limit its practical implementation, which is another obstacle for the hydrogen economy. We must once again seek for methods to make it safer.
- The most common method of producing hydrogen is through the reforming of fossil fuels. This procedure, on the other hand, necessitates a significant amount of energy and the discharge of hazardous substances into the environment. Fortunately, there are a variety of additional ways to create hydrogen.

SAMPLE TOPIC ACTIVITIES

Activity 1: Video Learning

Title: Hydrogen Economy

Instruction to Teacher:

1. Play a video that shows the students about how a Hydrogen Economy would look like. (Link of Sample Video: https://youtu.be/xDzHJ_a8LwY)
2. Ask questions to arouse their interest in Hydrogen Economy and how it can be adopted into Papua New Guinea.
3. Do you think hydrogen economy is possible in PNG?

Activity 2: Exercise

Complete the following questions:

1. Explain what Hydrogen Economy is?
2. Explain what Hydrocarbon Economy is?
3. Define Energy Density.
4. Why is Hydrogen an Energy Carrier?
5. Why do you think it is important for us to learn about Hydrogen Economy?
6. Do you think Papua New Guinea can build a Hydrogen Economy System?
7. How would a Hydrogen Economy revolutionize the current energy infrastructure of Papua New Guinea?
8. Papua New Guinea is rich with both non-renewable and renewable energy resources, how would you design a national energy system that integrates them both and protects the environment? (Explain your answers in General)

Activity 3: Group Work/Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research innovative ways in which they would integrate the concept of Hydrogen Economy into Papua New Guinea's Economy. Students must justify how a Hydrogen economy will benefit Papua New Guinea.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Topic 2: Methods of Producing Hydrogen

We mentioned that the key elements of a hydrogen economy are:

- **Production of Hydrogen**
- Packaging of Hydrogen
- Transportation of Hydrogen
- Storage of Hydrogen
- Transfer of Hydrogen
- End-users

In this topic, we will be discussing the first element: Production of Hydrogen. It is important that we become familiar with the different types of known processes that produces hydrogen. This knowledge may enable you to think of a way to help transition PNG into a green energy economy.

2.1 Color-coding Hydrogen Production Methods.

You've probably noticed that we've been referring to the hydrogen economy as a "green" one. The color "Green" was chosen to represent that form of economy not because hydrogen is green in color, but because the processes that produce hydrogen do not pollute the environment.

Colors have been assigned to different techniques of hydrogen generation by scientists. The primary colors we'll be discussing are:

- **Green Hydrogen** – hydrogen produced by electrolysis using electricity provided from renewable sources like hydro, solar, or wind. As a result, it produces no greenhouse gases, making it a great carbon-free fuel source. Water electrolysis, which uses an electrical current to divide water into hydrogen and oxygen components, produces green hydrogen. Water is released as a waste product when hydrogen is mixed with air to produce electricity.
- **Blue Hydrogen** – refers to hydrogen created by technologies like Steam Methane Reforming (SMR), but with the CO₂ by-product captured and stored during the manufacturing process. In general, carbon capture and storage means injecting CO₂ underground. Although it still releases greenhouse gases, blue hydrogen operations might potentially reduce the amount of carbon produced, making it a cleaner alternative to brown hydrogen.
- **Turquoise hydrogen** – is similar to blue hydrogen in that it uses a process called methane pyrolysis to produce hydrogen and solid carbon from methane, which may then be permanently stored and used.
- **Grey Hydrogen** – This is the name given to almost all hydrogen produced for industrial purposes. This means that the majority of hydrogen is made from natural gas using a process called Steam Methane Reforming (SMR), which involves mixing natural gas with water (as steam) to create hydrogen gas. In addition, the method emits a significant amount of CO₂ (about 9 parts for every one part of hydrogen it creates).

- **Brown (or Black) Hydrogen** – Thermochemical processes such as Coal Gasification from coal or lignite burning produce hydrogen. It produces the same amount of CO₂ as burning the source fuel would have done in the first place, making it the least environmentally friendly option. For every ton of brown hydrogen generated, around 10-12 tons of CO₂ are produced.
- In conclusion, Blue, Turquoise, Grey, Brown, or Black Hydrogen all refer to hydrogen that is produced via nonrenewable energy sources and so contributes to pollution. Green hydrogen, on the other hand, is hydrogen that is produced using renewable energy sources and hence is environmentally friendly.

The many methods of creating hydrogen will be discussed in the following sections. You can figure out which colors go well with certain production methods. The following is a summary of the discussion:

- Thermochemical Processes
 - Steam Methane Reforming (SMR)
 - Biomass Gasification
 - Biomass-derived Liquid Reforming
 - Thermochemical Water Splitting
- Electrolytic Processes
 - Water Electrolysis

2.2 Thermochemical Processes

Some thermal processes use energy from diverse sources, such as natural gas, coal, or biomass, to release hydrogen from its molecular structure. In other processes, heat is combined with closed-chemical cycles to produce hydrogen from feedstocks such as water.

2.2.1 Natural Gas Reforming

Natural gas reforming is a well-known and sophisticated industrial technology that makes use of existing natural gas pipeline infrastructure. Because it is a hydrogen-generating process, it is a critical technology for near-term hydrogen generation.

What is the mechanism behind it?

Methane (CH₄) is found in natural gas, and it can be transformed to hydrogen utilizing thermal processes including steam-methane reformation and partial oxidation.

2.2.2 Steam Methane Reforming (SMR)

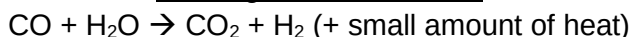
Steam methane reforming, a well-established industrial process that uses high-temperature steam (700°C-1000°C) to produce hydrogen from a methane source such as natural gas, can be used to make hydrogen. By reacting methane with steam at 3-25 pressure (1 bar = 14.5 psi) in the presence of a catalyst, SMR creates hydrogen (H), carbon monoxide (CO), and a trace amount of carbon dioxide (CO₂). Steam reformation is an endothermic reaction, which means it requires heat to continue.

In a process known as the "water-gas shift reaction," CO and steam are reacted with a catalyst to produce CO and extra H. In a final process stage termed as "pressure-swing absorption," carbon dioxide (CO₂) and other impurities are removed from the gas stream, leaving essentially pure hydrogen. Steam reforming may convert other fuels, such as ethanol, propane, or even gasoline, to hydrogen.

SMR Reaction



Water-gas Shift Reaction



2.2.3 Partial Oxidation

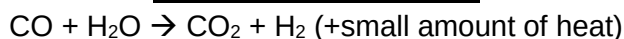
When methane and other hydrocarbons in natural gas react with a small amount of oxygen (typically from the air), the hydrocarbons are partially oxidized, resulting in carbon dioxide and water. When there isn't enough oxygen, the reaction produces predominantly hydrogen and carbon monoxide (and nitrogen if the procedure is done with air rather than pure oxygen), with a trace of CO₂ and other molecules. In a water-gas shift process, CO is combined with water to produce CO₂ and extra hydrogen.

The partial oxidation process is exothermic, which means it generates heat. The process is usually significantly faster than steam reforming, and it only requires a small reactor vessel. This approach initially provides less hydrogen per unit of input fuel than steam reforming the same fuel, as may be seen in partial oxidation chemical reactions.³

Partial Oxidation of Methane Reaction



Water-gas Shift Reaction



2.2.4 Biomass Gasification

Biomass gasification is a well-established technical pathway that uses heat, steam, and oxygen to transform biomass into hydrogen and other products without burning it. Net carbon emissions from this strategy can be minimal since biomass growth removes carbon dioxide from the atmosphere, especially when paired with long-term carbon collection, utilization, and storage. The development and operation of biofuel gasification facilities is underway, and best practices and lessons learned for hydrogen production can be implemented.

2.2.4.1 What is Biomass?

Biomass is a renewable organic resource that includes agriculture crop residues (such as maize stover or wheat straw), forest residues, special crops designed specifically for energy use (such

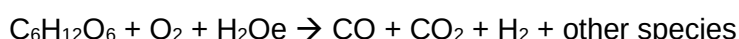
³ (Hydrogen Production: Natural Gas Reforming, 2021)

as switchgrass or willow trees), organic municipal solid trash, and animal wastes. From this sustainable resource, gasification can be used to produce hydrogen as well as other byproducts.

2.2.4.2 How Does Biomass Gasification Work?

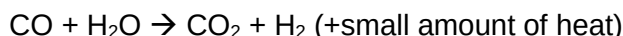
Gasification is a process that converts organic or fossil-based carbonaceous materials into carbon monoxide, hydrogen, and carbon dioxide at high temperatures (>700°C) with a controlled amount of oxygen and/or steam. In a water-gas shift process, carbon monoxide reacts with water to produce carbon dioxide and additional hydrogen. To extract hydrogen from this gas stream, adsorbers or specific membranes can be utilized.

Simplified example Reaction



NOTE: The above process substitutes glucose for cellulose. The content and complexity of actual biomass varies greatly, with cellulose being one of the most important components.

Water-gas Shift Reaction



Pyrolysis is the process of gasifying biomass in the absence of oxygen. Biomass does not gasify as easily as coal, resulting in the formation of extra hydrocarbon compounds in the gas mixture entering the gasifier, particularly when no oxygen is used. As a result, a step of reforming these hydrocarbons with a catalyst to produce a clean syngas mixture of hydrogen, carbon monoxide, and carbon dioxide is normally necessary. Similar to the gasification process for hydrogen production, the carbon monoxide is subsequently transformed to carbon dioxide in a shift reaction step (with steam). The hydrogen is then isolated and purified.⁴

2.2.5 Biomass-derived Liquid Reforming

Liquids derived from biomass resources, such as ethanol and bio-oils, can be reformed to produce hydrogen in a process similar to natural gas reforming. Biomass-derived liquids are more transportable than biomass feedstocks, allowing enabling semi-central or even distributed hydrogen generation at filling stations. Liquid reforming from biomass is a technique that will be available in the mid-term.

2.2.5.1 How Does it Work?

Biomass resources can be used to produce cellulosic ethanol, bio-oils, and other liquid biofuels. Some of these liquids may be transported for free to a refueling station or another location of use, where they could be reformed into hydrogen. Others, such as bio-oils, might be produced on-site.

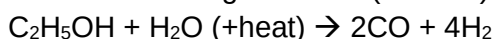
The process of reforming biomass-derived liquids to hydrogen, which is comparable to natural gas reforming, includes the following steps:⁵

⁴ (Hydrogen Production: Biomass Gasification, 2021)

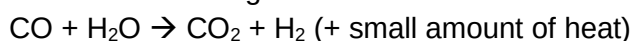
⁵ (Hydrogen Production: Biomass Derived Liquid Reforming, 2021)

- (1) In the presence of a catalyst, the liquid fuel is mixed with steam at high temperatures to produce reformat gas, which is predominantly hydrogen, carbon monoxide, and a minor amount of carbon dioxide.
- (2) By combining carbon monoxide (produced in the first stage) with high-temperature steam, the "water-gas shift reaction" produces more hydrogen and carbon dioxide.
- (3) The hydrogen is finally purified and separated.

Steam Reforming Reaction (Ethanol)



Water-gas Shift Reaction



2.2.6 Thermochemical Water Splitting

Thermochemical water splitting combines high temperatures (from concentrated solar power or waste heat from nuclear power operations) with chemical processes to produce hydrogen and oxygen from water. This is a long-term technical path that could result in low or no greenhouse gas emissions in the future.

2.2.6.1 What is the Process?

In thermochemical water splitting systems, high-temperature heat ($500^{\circ}\text{--}2,000^{\circ}\text{C}$) is utilized to power a series of chemical processes that produce hydrogen. Each cycle recycles the chemicals used in the process, creating a closed loop that creates hydrogen and oxygen using only water. High temperatures can be obtained in a number of ways, including directing sunlight onto a reactor tower using a field of mirror "heliostats," as seen in Figure SC-E-SRB-2.

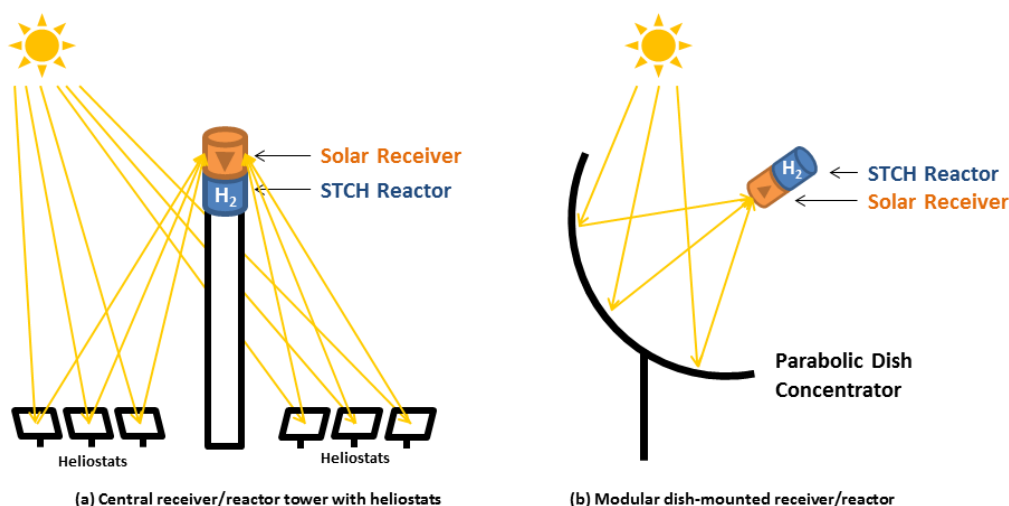


Figure SC-E-SRB- 2: (a) Central Receiver/Reactor Tower with Heliostats (b) Modular dish-mounted receiver/reactor
(Source: [Hydrogen Production: Thermochemical Water Splitting | Department of Energy](#))

For hydrogen production, a variety of solar thermochemical water-splitting cycles have been researched, each with its own set of operating parameters, engineering constraints, and hydrogen production prospects.

The "direct" two-step cerium oxide thermal cycle and the "hybrid" copper chloride cycle are two examples of thermochemical water splitting cycles shown in Figure SC- E-SRB-3. Typically, direct cycles are less complex with fewer steps, but they require higher operating temperatures compared with the more complicated hybrid cycles.⁶

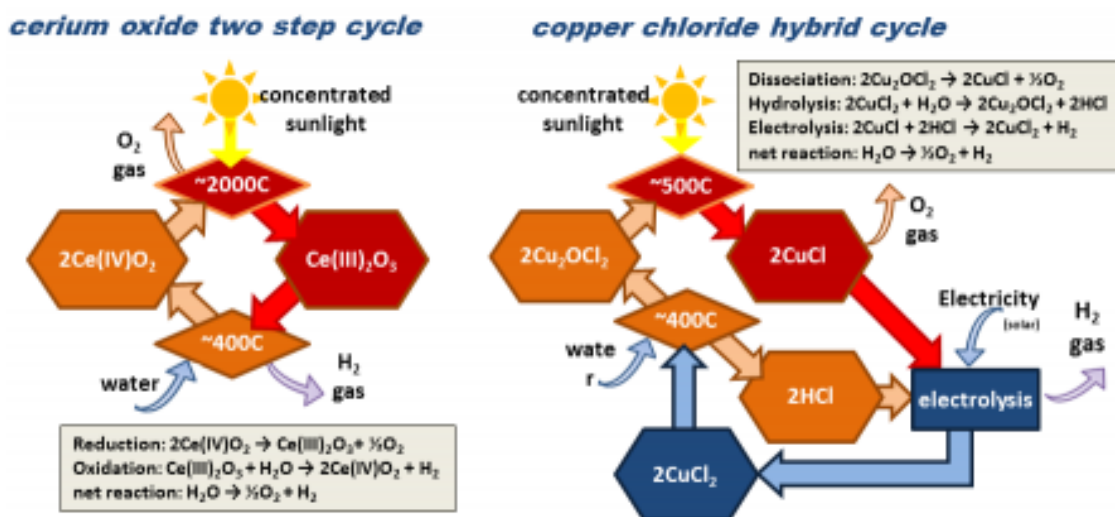


Figure SC-E-SRB- 3: Cerium oxide two step cycle (left) and copper chloride hybride cycle (right) (source: [Hydrogen Production: Thermochemical Water Splitting | Department of Energy](#))

2.3 Electrolytic Processes

From renewable and nuclear energy, electrolysis is a potential way for producing carbon-free hydrogen. Electrolysis is the use of electricity to split water into hydrogen and oxygen. This reaction is carried out in an electrolyzer. Electrolyzers can be small, appliance-sized equipment that is well-suited for small-scale dispersed hydrogen production to large-scale, central production facilities that are directly linked to renewable or other non-greenhouse-gas-emitting sources of power supply.

2.3.1 How does it work?

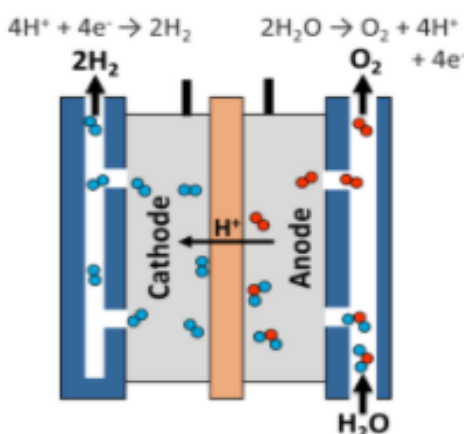


Figure SC-E-SRB- 4: Hydrogen Fuel Cell (Source: [Hydrogen Production: Photoelectrochemical Water Splitting | Department of Energy](#))

⁶ (Hydrogen Production: Thermochemical Water Splitting, 2021)

2.3.2 Types of Electrolyzers

Electrolyzers have an anode and a cathode separated by an electrolyte, similar to fuel cells. Different electrolyzers work in different ways due to the various types of electrolyte material and ionic species they conduct.

2.3.2.1 Polymer Electrolyte Membrane Electrolyzers

A solid specialized plastic substance serves as the electrolyte in a polymer electrolyte membrane (PEM) electrolyzer. Water reacts with oxygen at the anode to produce oxygen and positively charged hydrogen ions (protons). As electrons flow through an external connection, hydrogen ions preferentially travel across the PEM to the cathode. At the cathode, hydrogen ions combine with electrons from the external circuit to produce hydrogen gas.

Anode Reaction: $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$

Cathode Reaction: $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2$

2.3.2.2 Alkaline Electrolyzers

Hydroxide ions (OH^-) are transported from the cathode to the anode through the electrolyte, with hydrogen being produced on the cathode side. Electrolyzers that use a liquid alkaline sodium or potassium hydroxide solution as the electrolyte have been around for quite some time. Newer approaches using solid alkaline exchange membranes (AEM) as the electrolyte are showing promise in the lab.

2.3.2.3 Solid Oxide Electrolyzers

Solid oxide electrolyzers create hydrogen in a different way, using a solid ceramic material as the electrolyte and selectively conducting negatively charged oxygen ions (O^{2-}) at high temperatures. Steam reacts with electrons from the external circuit at the cathode to form hydrogen gas and negatively charged oxygen ions, which then flow through the solid ceramic membrane and react with oxygen gas at the anode to produce electrons for the external circuit.

Solid oxide electrolyzers must operate at temperatures high enough for solid oxide membranes to function properly (about $700^\circ\text{--}800^\circ\text{C}$, compared to PEM electrolyzers at $70^\circ\text{--}90^\circ\text{C}$ and commercial alkaline electrolyzers at less than 100°C). Advanced lab-scale solid oxide electrolyzers with proton-conducting ceramic electrolytes show potential in decreasing the working temperature to $500^\circ\text{--}600^\circ\text{C}$. The solid oxide electrolyzers can exploit the heat available at these high temperatures (from a variety of sources, including nuclear energy) to reduce the amount of electrical energy required to create hydrogen from water.

2.4 Direct Solar-Water Splitting Processes

Photolytic techniques, also known as direct solar water splitting, use light energy to split water into hydrogen and oxygen. These technologies are currently in varying phases of development, but they have the potential to produce sustainable hydrogen with minimal environmental impact in the long run.

2.4.1 Photoelectrochemical (PEC)

Hydrogen is created from water via photoelectrochemical (PEC) water splitting, which uses sunlight and specialized semiconductors called photoelectrochemical materials to directly separate water molecules into hydrogen and oxygen. This is a long-term technology route that has the potential to emit very little or no greenhouse gases.

2.4.1.1 How does it work?

The PEC water splitting process converts solar energy directly to chemical energy in the form of hydrogen using semiconductor materials. The semiconductor materials used in the PEC process are similar to those used in photovoltaic solar power generation, but the semiconductor is immersed in a water-based electrolyte for PEC applications, where sunlight energizes the water-splitting process. At the National Renewable Energy Laboratory, you may see a laboratory-scale demonstration of this process.

PEC reactors are available in panel form (similar to photovoltaic panels), as electrode systems, or as slurry-based particle systems, each with its own set of benefits and drawbacks. Panel systems have received the greatest attention to date, owing to their parallels to existing photovoltaic panel technology. To see some various possible implementations of both the panel and slurry reactor concepts, click on the title of each figure.

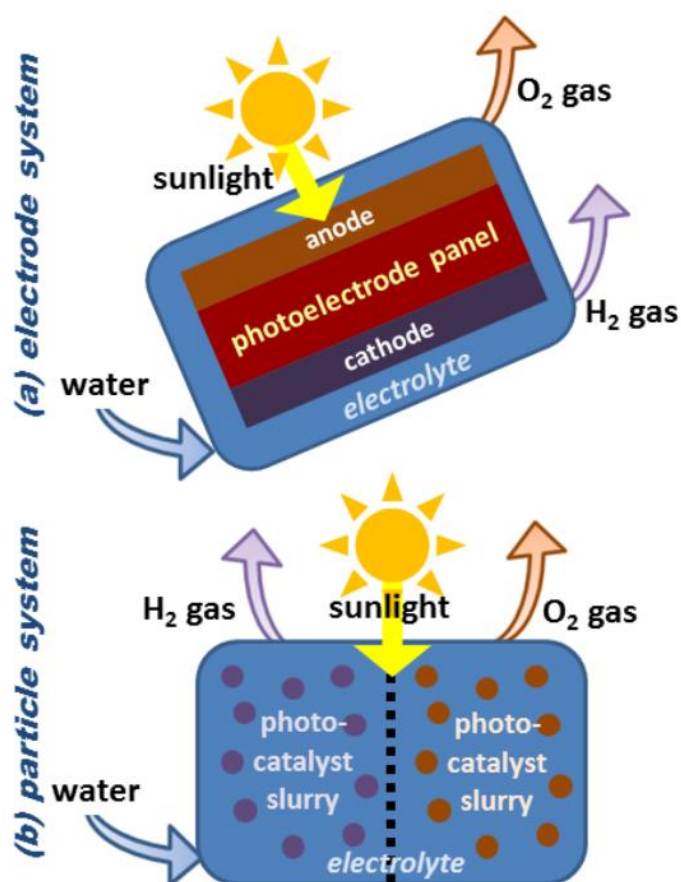


Figure SC-E-SRB- 5: Two different approaches to PEC solar hydrogen production reactors. (a) electrode systems similar to flat-plate photovoltaic panels; and (b) particle systems comprised of slurries of PEC semiconductor particles. Source. [Hydrogen Production: Photoelectrochemical Water Splitting | Department of Energy](#)

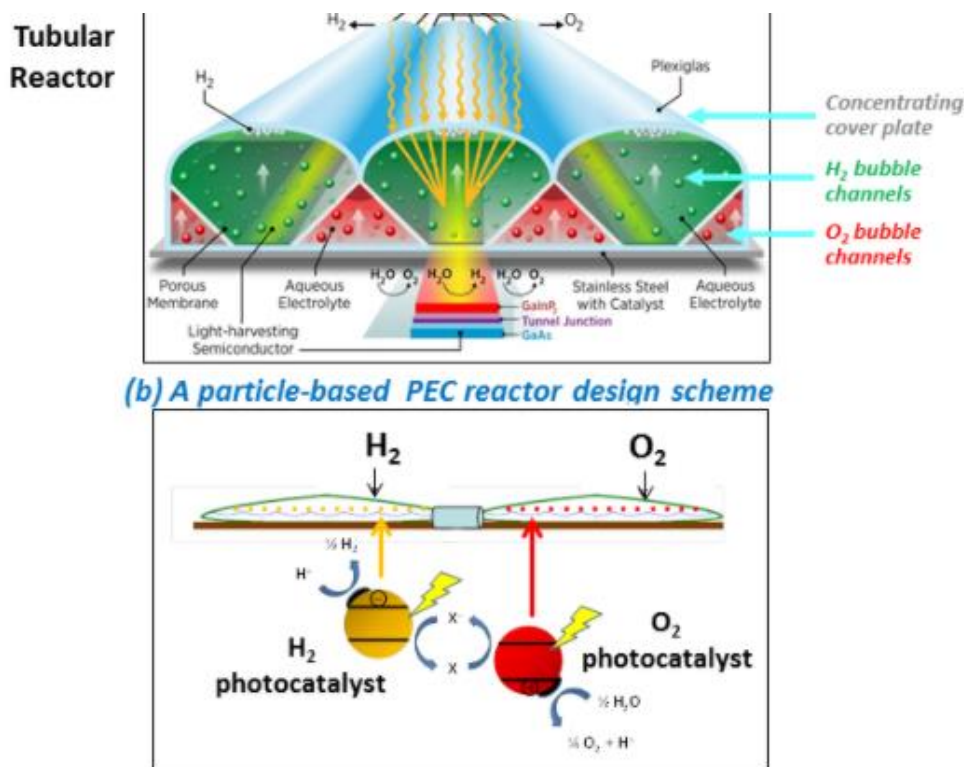


Figure SC-E-SRB- 6: Possible PEC reactor design scheme for (a) electrode systems, including a flat plate and a tubular reactor (providing moderate solar concentration onto one electrode strip); and (b) a plastic "baggie" covered dual bed particle reactor with wide-by-side photocatalyst slurries. Source. [Hydrogen Production: Photoelectrochemical Water Splitting | Department of Energy](#)

2.4.1.2 Why is this Pathway Being Considered?

PEC water splitting is a promising solar-to-hydrogen method, with high conversion efficiency at low working temperatures and the use of cost-effective thin-film and/or particle semiconductor materials.

2.5 Where does the hydrogen come from?

The hydrogen itself is one of the most intriguing aspects of the hydrogen economy. Where is it going to come from? You simply pump the fossil fuel out of the ground and refine it in the fossil fuel economy. Then you burn it as a source of energy.

Oil, gasoline, coal, and natural gas are all taken for granted by most people, but they are truly extraordinary. These fossil fuels were sun energy that had been stored for millions of years. Plants have used sun energy to power their growth for millions of years. They finally died and were transformed into oil, coal, and natural gas. We 'freely' tap into that vast solar energy reservoir when we extract oil from the earth. We unleash that stored solar energy every time we consume a gallon of gasoline.

There is no storage to tap into in the hydrogen economy. We need to be able to generate energy in real time. There are two conceivable hydrogen sources:

- Electrolysis of Water** – It is simple to split water molecules using electricity to produce pure hydrogen and oxygen. One of the biggest advantages of this method is that it may be carried

out anyplace. For instance, you could have a box in your yard that produces hydrogen from tap water, and you could use that hydrogen to fuel your car.

- b. **Reforming Fossil Fuels** – Hydrocarbon molecules, which are made up of hydrogen and carbon, are found in oil and natural gas. You may simply slit the hydrogen off the carbon in a hydrocarbon using a device called a "Fuel Processor" or a "Reformer," and then use the hydrogen. You release the remaining carbon as carbon dioxide into the atmosphere.

NOTE: We will discuss the methods of producing hydrogen in Topic 2 of this Module.

Of course, the second option is a little strange. For the hydrogen economy, you're relying on fossil fuels as a source of hydrogen. This method minimizes air pollution, but it does not address the greenhouse gas problem (since carbon is still emitted into the environment) or the dependency issue (you still need oil). However, it could be a useful interim step in the transition to a hydrogen economy.

When you hear about "fuel-cell-powered automobiles" being developed by automotive companies right now, you'll notice that practically all of them aim to get hydrogen for the fuel cells by reforming gasoline. The reason for this is because gasoline is a readily available hydrogen source. This is the simplest way to collect hydrogen to power a vehicle's fuel cell until "hydrogen stations" appear on every corner like gas stations do now.

The first option is intriguing since it is at the heart of the true hydrogen economy. To achieve a pure or "Green Hydrogen Economy," hydrogen must be generated from renewable sources rather than fossil fuels, as this will prevent carbon from being released into the atmosphere. The largest change in the hydrogen economy will be having enough electricity to extract hydrogen from water and generating that electricity without using fossil fuels.

2.6 Where will the electricity for the electrolysis of water come from?

Currently, coal or natural gas are used to generate almost all of the world's electricity (non-renewable energy sources). In the hydrogen economy, all of that generating capacity will have to be replaced by renewable sources. Furthermore, all fossil fuel energy currently utilized for transportation (in cars, trucks, trains, boats, and planes) will have to be converted to hydrogen, which will be produced using electricity. To put it another way, a country's electrical generating capacity will have to double to meet transportation demands, and then it will all have to switch from fossil fuels to renewable sources. At that point, and only at that point, will the flow of carbon into the atmosphere stop.

Right now, there are several different ways to create electricity that do not use fossil fuels:

- Nuclear Power
- Hydroelectric Dams
- Solar Cells
- Wind Turbines
- Geothermal Power
- Wave and Tidal Power

- Co-generation – For example, a sawmill might burn bark to create power, or a landfill might burn methane that the rotting trash produces).⁷

In **Papua New Guinea**, about 20% of the power currently comes from nuclear and 7% comes from hydroelectric. Solar, wind, geothermal and other sources generate only 5 percent of the power – hardly enough to matter.

In the future, barring some technological breakthrough, it seems likely that replacement of fossil fuel power plants will be replaced by renewable energy power plants to create the hydrogen economy. Remember that, in a Green Hydrogen Economy, the electrical generating capacity will have to approximately double because all of the energy for transportation that currently comes from oil will have to be replaced with electrically generated hydrogen. So, the number of power plants will double, and all of the fossil fuel plants will be replaced.

The electrical-generation problem is probably the biggest barrier to overcome for the hydrogen economy. Once the technology is refined and becomes inexpensive, fuel-cell vehicles powered by hydrogen could replace gasoline internal combustion engine vehicles (ICEVs) over the course of a decade or two.

2.6.1 How do you store and transport the hydrogen?

The challenge with putting pure-hydrogen automobiles on the road right now is storage and transportation. Hydrogen is a bulky gas that is far more difficult to work with than gasoline. Compressing the gas takes energy, although compressed hydrogen has significantly less energy than gasoline of the same volume. However, answers to the challenge of hydrogen storage are emerging.

For example, a chemical known as "Sodium Borohydride" can be used to store hydrogen in a solid form. Chrysler is testing this technology, so it's been in the headlines recently. Borax is used to make this chemical (a common ingredient in some detergents). As the hydrogen in sodium borohydride is released, it decomposes into borax, which can then be recycled.

Once the storage problem has been solved and standardized, a network of hydrogen stations as well as transportation infrastructure will need to emerge to support it. The technological sorting-out process may be the main impediment. Stations will take a long time to develop until a storage technology clearly dominates the market. For example, if all hydrogen-powered automobiles from all manufacturers used solidum borohydride, a station network could emerge swiftly; however, if history is any indicator, such standardization is unlikely to occur quickly.

There could also be a technical breakthrough that drastically alters the game. Electric cars, for example, would not require fuel cells and would not require hydrogen on the road if someone could produce an inexpensive rechargeable battery with a high capacity and rapid recharging time. Electricity would be used to directly refuel automobiles.

⁷ (Brain, 2021)

2.6.2 Prospects for the Future

Because hydrogen is the future of energy, you will hear more and more about it in the news. The environmental challenges associated with the fossil fuel economy are colliding with advances in fuel-cell technology, allowing us to take the first steps.

The marketing of fuel-cell-powered vehicles will be the most visible phase. Fuel cells have two important advantages over internal combustion engines, despite the fact that they will initially be powered by gasoline and reformers.

- They are about twice as efficient
- They can significantly reduce air pollution in cities.

Gasoline-powered fuel-cell vehicles are an excellent transitional step because of those advantages.

It will be difficult to transition to a Green Hydrogen Economy. Power plants will have to move to renewable energy sources, and the market will have to agree on methods for storing and transporting hydrogen. Due to these roadblocks, the shift to a hydrogen economy will most certainly take a long time.⁸

⁸ (Brain, 2021)

SAMPLE TOPIC ACTIVITIES

Activity 1: Exercise

- (1) List and explain the different Thermochemical Processes to produce Hydrogen
- (2) List and explain the different Electrolytic Processes to produce Hydrogen
- (3) Explain how Direct Solar-Water Splitting process is used to produce Hydrogen

Activity 2: Practical Assignment/Experiment

Title: Electrolyzing Water

Objective(s): To produce hydrogen

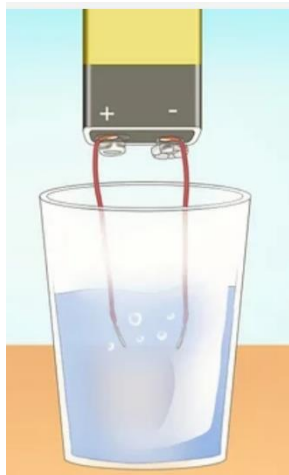
Background

Both water and acids include hydrogen, which may be separated to produce pure hydrogen gas. Electricity may be used to extract hydrogen gas from water. Many acids will react with metals such as aluminum to produce hydrogen gas, making them even easier to employ. Just keep in mind that hydrogen gas can cause an explosion.

Procedure

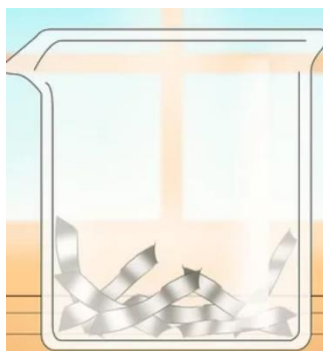
Part A: Electrolyzing Water

1. Pour water into a glass container - Avoid using plastic or metal for this. You will be running an electric current through the water that could melt plastics. Metal will conduct this current and could shock you if you touch the container
 - i. If you are unfamiliar with this experiment, start with one cup of water. Producing too much hydrogen can be dangerous.
 - ii. Children should only do this experiment with a responsible/knowledgeable adult.
 - iii. Add a tablespoon of salt to the water for better results — salt will help conduct the electrical current.
2. Locate Power Source – While large-scale hydrogen synthesis necessitates a significant amount of energy, this experiment may be carried out with little resources. The electrolysis may be done using a 9V battery (splitting of the water into hydrogen and oxygen). To increase the power of the effect, connect more batteries in series.
 - i. A portable power supply is good for this experiment, as this experiment should be done outside or under a fume hood.
 - ii. While batteries aren't generally dangerous, you should still wear rubber gloves to avoid being shocked.
3. Connect a paper clip to each terminal of the battery - This will produce an anode (the negative paper clip) and a cathode (the positive paper clip) (the positive paper clip). Wrap the paper clips around the battery connections until they're secure.
4. Submerge the paper clips – When you immerse the paper clips in water, electricity will flow from the anode to the cathode. The anode will create hydrogen bubbles, while the cathode will produce oxygen and chlorine gas.
 - i. Be cautious not to touch the paper clips together.



Part B: Reacting to Acid

1. Place Aluminum foil in a beaker - Tear aluminum foil into small pieces and place them in a beaker or other container. A container with a lid or cap should not be used, since this might result in an explosion. The amount of aluminum utilized does not need to be measured.



- i. A three-inch-by-three-inch square of aluminum foil may be ripped into bits the size of your thumb.
2. Keep Beaker Ventilated - Experiment either outside or under a fume hood. Hydrogen is incredibly combustible and disperses fast. An accumulation of hydrogen gas exposed to the air (or any other source of oxygen) has the potential to explode.
 - i. One example of hydrogen gas exploding is the Hindenburg.
 3. Add hydrochloric Acid - The amount of hydrochloric acid used does not need to be accurately measured. The metal will react with the chlorine in the hydrochloric acid to generate aluminum chloride. As a consequence, hydrogen gas will be produced.
 - i. Start with about two ounces of hydrochloric acid and add more if needed.

Part C: Collecting the Hydrogen



1. Use a balloon or bottle to collect the hydrogen - Place the collecting vessel's (bottle or balloon) aperture over the reaction beaker or container's opening. Because hydrogen gas has a lower density than air, it will rise to the top of your collecting jar. This approach is frequently used to show the upward flow of hydrogen gas by inflating a balloon.
2. Avoid collecting hydrogen when both the paper clips are releasing gas - If hydrogen gas comes into touch with oxygen, it can explode. If you're electrolyzing, don't try to gather hydrogen while the paper clips are still generating gas since the gases will be a mix of hydrogen and oxygen.
3. Collect only small amount - A large amount of hydrogen gas should not be collected or stored. It presents a significant threat to public safety and is difficult to contain. This experiment should only be carried out for the sake of amusement or instruction.

Results and Discussion

Discuss and present the results of the experiment.

Conclusion

Conclude with an explanation about how hydrogen is produced via electrolysis.

Activity 3: Group Work/Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research on a method of producing Hydrogen that's of there. Students must come up with innovative ideas to use that method in modern application.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Topic 3: Storing Hydrogen

3.1 How do you store and transport the hydrogen?

The challenge with putting pure-hydrogen automobiles on the road right now is storage and transportation. Hydrogen is a bulky gas that is far more difficult to work with than gasoline. Compressing the gas takes energy, although compressed hydrogen has significantly less energy than gasoline of the same volume. However, answers to the challenge of hydrogen storage are emerging.

For example, a chemical known as "Sodium Borohydride" can be used to store hydrogen in a solid form. Chrysler is testing this technology, so it's been in the headlines recently. Borax is used to make this chemical (a common ingredient in some detergents). As the hydrogen in sodium borohydride is released, it decomposes into borax, which can then be recycled.

Because hydrogen is the future of energy, you will hear more and more about it in the news. The environmental challenges associated with the fossil fuel economy are colliding with advances in fuel-cell technology, allowing us to take the first steps.

There could also be a technical breakthrough that drastically alters the game. Electric cars, for example, would not require fuel cells and would not require hydrogen on the road if someone could produce an inexpensive rechargeable battery with a high capacity and rapid recharging time. Electricity would be used to directly refuel automobiles.

3.2 How Does it Work / is it Stored?

Hydrogen can be stored in three different ways:

1. As a gas under high pressures
2. In liquid form under cryogenic temperatures
3. On the surface of or within solid and liquid materials

Each of these storage techniques has its own requirements and challenges, as shown below:

3.2.1 Compressed Gas

Under high pressures, hydrogen can be compressed and held as a gas. This necessitates storage tanks with pressures ranging from 350 to 700 bar (5000 to 10,000 psi).

3.2.1.1 Cryogenic Liquid Storage

Hydrogen can be held as a liquid cryogenically. Low temperatures are needed to prevent liquid hydrogen from boiling off and becoming a gas, which happens at -252.8°C . Although liquid hydrogen has a higher energy density than gaseous hydrogen, getting it to the needed temperatures can be expensive. Furthermore, cryogenic liquid hydrogen storage tanks and facilities must be insulated to prevent evaporation if any heat is transported into the liquid hydrogen by conduction, convection, or radiation. Despite these obstacles, liquid hydrogen is in considerable demand for applications that require high purity, and it is even employed in space flight.

3.2.1.2 Combined Cold-and-Cryo-Compressed Hydrogen

The above-mentioned compression and cryogenic cooling storage technologies can be coupled to build a more advanced hydrogen storage system. The hydrogen is cooled before being crushed in this case. This results in a higher energy density than compressed hydrogen, but it also requires more energy to achieve, similar to cryogenic liquid storage.

The energy required for these various methods of hydrogen storage accounts for 9-12 percent of the energy available for compression (from 1 to 350 or 700 bar) and roughly 30% for liquefaction. The amount of energy used varies depending on the particular method, amounts, and environmental conditions, although research is continuing to identify more cost-effective storage ways that use less energy.

3.3 Materials-Based Hydrogen Storage

Hydrogen can be kept in a variety of ways, including compressed as a gas or as a liquid. There are three types of hydrogen storage materials: those that store hydrogen on the surface of the material, those that store hydrogen within the material, and hydride storage, which employs a combination of solid and liquid materials.

Adsorption is the process of hydrogen molecules or atoms adhering to the surface of a substance. The hydrogen binds to materials having high surface areas, such as microporous organometallic framework complexes (MOFs), microporous crystalline aluminosilicates (zeolites), or microscopically small carbon nanotubes, in this process. Due to the larger surface area of the sorbent, hydrogen adsorption to powdered materials can attain significant densities of volumetric storage.

Hydrogen is split into hydrogen atoms during absorption and integrated into a material's internal solid lattice system.

The reaction of hydrogen-containing materials with water or other liquid compounds, such as alcohols, is used in hydride storage, the third of these material storage systems for hydrogen. The hydrogen is efficiently held in both the substance and the liquid in this type of hydrogen storage, also known as 'chemical hydrogen storage.'

Hydrogen forms an interstitial compound with elemental metals like palladium, magnesium, and lanthanum, intermetallic compounds, light metals like aluminum, or some alloys in metal hydride storage systems. These metal hydrides adsorb molecular hydrogen onto their surfaces and subsequently incorporate it into the metallic lattice in elemental form, producing heat. They may be re-released with heat, and these hydrides can absorb vast amounts of gas, with palladium, for example, absorbing 900 times its own volume of hydrogen.

A liquid organic hydrogen carrier can also be used to chemically bind hydrogen. These chemical compounds, which include the carbazole derivative N-ethylcarbazole and toluene, have a high potential for hydrogen absorption.

Large amounts of hydrogen may be held in materials with smaller volumes, lower pressures, and temperatures near to room temperature using these material-based approaches. Volumetric storage densities larger than those of liquid hydrogen are possible using materials-based storage.

Materials-based storage, on the other hand, is still in development since the cost of charging, discharging, and processing hydrogen is currently considered too costly and time-consuming.

3.3.1 Underground Hydrogen Storage

Salt caverns, depleted oil and gas fields, and aquifers can all be used to store hydrogen on a large scale. For years, underground storage sites for natural gas and crude oil have been utilized to balance supply and demand changes, as well as to prepare for a catastrophe.

The most expensive option is cavern storage, but it is also the most ideal for hydrogen storage. Only a few locations in Europe and the United States have operational experience with cavern-based hydrogen storage. Depleted underground natural gas stockpiles are the most prevalent of them, and they're employed as hydrogen reservoirs for surplus renewable energy.

3.3.2 Gas Grid Hydrogen Storage

Surplus hydrogen can be put into the public natural gas network to provide hydrogen enriched natural gas as an alternative to subterranean cavern storage (HENG).

Throughout the twentieth century, hydrogen-enriched town gas or coke-oven gas, with a hydrogen content of more than 50% by volume, was transported to residences in Germany, the United States, and the United Kingdom via gas pipelines. Although it was later upgraded to transmit natural gas, the infrastructure utilized at the time still stands.

While it is widely assumed that gas containing 10% hydrogen may be put into the existing natural gas system without harming end users or pipeline infrastructure, a number of crucial components have been declared unsuitable for usage at these hydrogen concentrations.

Despite this disadvantage, it is believed that substantial amounts of hydrogen gas might be stored in this manner by utilizing most of the existing natural gas networks in industrialized countries and then transformed directly into power using hydrogen fuel cells.

3.4 Why Hydrogen Storage is Important

If hydrogen is to be a part of the future renewable energy mix, it must be stored. Hydrogen fuel cells could help offer a greener solution to our power generating demands, including powering everything from small electronic gadgets to vehicles, aircraft, and even entire buildings, as part of international efforts to cut emissions and the usage of carbon-based fuels.

Another benefit of hydrogen as an energy source is that it may be created by electrolysis from excess renewable energy electricity, allowing hydrogen to meet a matching energy demand. Hydrogen, on the other hand, can be stored in enormous quantities for long periods of time. Because, unlike batteries, this energy does not degrade over time, it may be produced and stored on a large scale as part of a green energy mix. When needed, the hydrogen can be collected and used as a backup energy source.

In the transportation industry, hydrogen can be employed as a complementary fuel source alongside batteries. The hydrogen system will store the majority of the energy, while a modest

capacity battery will operate as a buffer to offer regenerative braking, meet any unexpectedly high power needs, and extend the lifetime of hydrogen fuel cells by reacting to load fluctuations. Some commercially available vehicles, such as the Honda FCX Clarity hydrogen car, already use this additional fuel technology. Of course, hydrogen fuel cells have been safely used for decades to supply clean electricity for forklifts that must operate in confined spaces.

Hydrogen storage is critical if hydrogen is to be a component of our future clean energy solutions; nevertheless, further research and infrastructure upgrades are needed for hydrogen to reach its full potential. As a result, the Hydrogen and Fuel Cell Technologies Office (HFTO) of the United States focuses on applied research, development, and innovation to expand hydrogen utilization for transportation and a variety of other purposes. Meanwhile, the US Department of Energy (DOE) funds research and development of a variety of technologies for producing hydrogen that is both cost-effective and ecologically beneficial.

3.5 Why is it Difficult to Store Hydrogen?

Because of its poor volumetric energy density, hydrogen is difficult to store. Because it is the lightest and simplest of all elements, it is quickly lost into the atmosphere, being lighter than helium.

Another issue is liquid hydrogen's extremely low boiling point (252.8°C), which necessitates cryogenically storing it at extremely low temperatures. Storing hydrogen as a gas has its own set of difficulties, as it usually necessitates the use of high-pressure containers (350-700 bar or 5000-10,000 psi).

As previously stated, another method for storing hydrogen is by adsorption or absorption; however, with these storage techniques, additional processes are required to release the hydrogen.

3.6 Is Hydrogen Storage Safe?

The risk level of all fuels is determined by three factors: the ignition source, the oxidant, and the presence of the fuel itself. Any fuel type, including hydrogen, may be made safer by using the right engineering controls.

In reality, hydrogen has a variety of characteristics that make it safer than many other types of fuel. It's non-toxic, for example, and because it's lighter than air, it dissipates swiftly when released into the atmosphere. This is significant because it means that in the event of an accident, the fuel will evaporate into the air rather than remaining in place and potentially catching fire, as is the case with batteries or petroleum, for example.

However, there are still risks associated with hydrogen, necessitating the implementation of additional engineering controls to assure its safe use. Because hydrogen has a lower ignition energy than gasoline or natural gas, it has a wide range of dangerous concentrations in the air, necessitating ventilation and leak detection in hydrogen systems. Because hydrogen burns with a nearly-invisible flame, special flame detectors are also necessary. Material selection is especially crucial for hydrogen systems since some metals become brittle when exposed to hydrogen.

Staff should be trained in how to securely handle hydrogen, and systems should be inspected for leaks and other possible issues to ensure it is produced, stored, and delivered safely. Of course, we've already seen hydrogen utilized in a wide range of everyday applications, demonstrating that we can improve safety and develop confidence in hydrogen as a future safe, clean, and renewable fuel.⁹

⁹ [What is Hydrogen Storage and How Does it Work? - TWI \(twi-global.com\)](https://www.twi-global.com/technical-studies/articles-view.aspx?article=112)

SAMPLE TOPIC ACTIVITIES

Activity 1: Group Work/Presentation

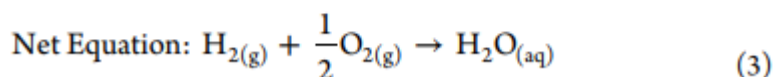
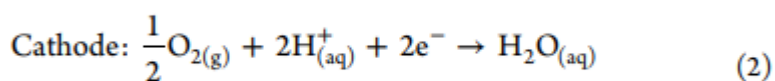
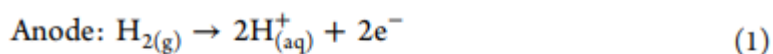
Instruction to Teachers

1. Group students
2. Task Groups to research on innovative ways that is of their interest to store hydrogen and justify how it is safe and applicable.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 2: Practical (optional)

Introduction

Meeting energy needs in a sustainable and environmentally responsible way is currently a major global challenge. Some strategies to address this problem are to involve very efficient uses of energy as well as renewable and sustainable (carbon-neutral) energy sources. A fuel cell converts chemical energy to electric energy with high efficiency. Often hydrogen is the energy storage medium in the fuel cell and oxygen the oxidizer, as shown in the abstract image and in eqs 1–3, and only the environmentally friendly water is released as the byproduct.



If hydrogen is used on a large-scale basis, the storage of hydrogen becomes a crucial issue for mobility and transport applications. Fuel cell technology would be practical if hydrogen could be stored in a safe, efficient, compact, and economic manner.² Since many compounds contain hydrogen, the goals of these experiments are to measure the hydrogen storage capacity of: (1) aqueous hydrochloric acid solution via reaction with Zn metal and (2) solid magnesium hydride by hydrolysis in the presence of acetic acid using an ideal gas law apparatus for hydrogen yield measurement and consumption of hydrogen by a fuel cell. Metal hydrides are advantageous for storage of hydrogen in a solid compound because high volumes of hydrogen are contained in small masses of metal hydride.

These experiments illustrate fundamental principles for handling gases and important gas law equations, offering direct experience with a hydrogen fuel cell and evaluating its performance, and introducing the concept of hydrogen storage materials, which is a central issue in the development of hydrogen economy. The total time for the experiment is about 2 h including prelab discussion, equipment setup, and conducting the experiment.

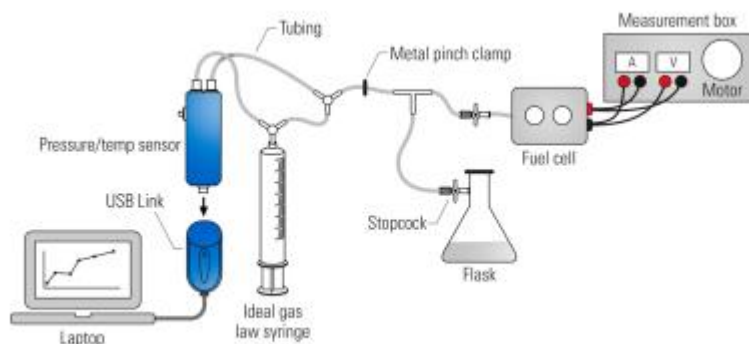


Figure 1. Apparatus for measuring the amount of produced hydrogen with fuel cell, motor, ideal gas law equipment, and laptop computer.

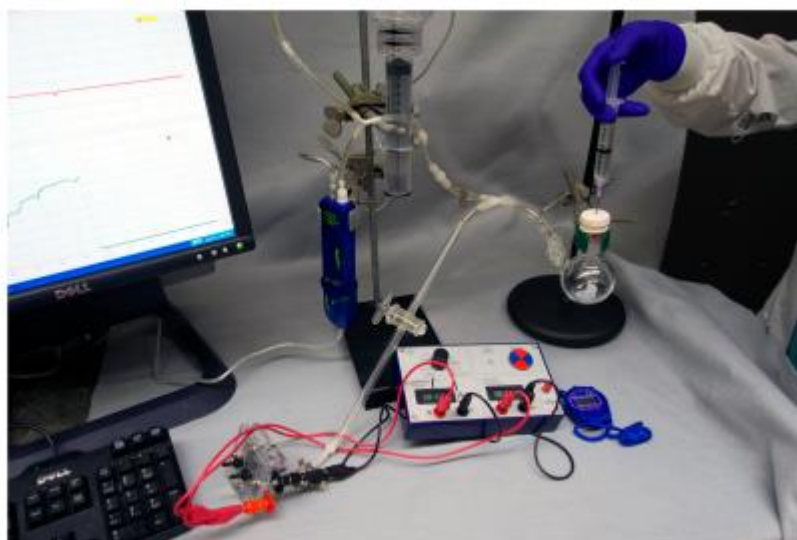
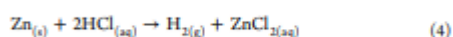


Figure 2. Experimental setup of equipment for determining hydrogen storage capacity of hydrogen-containing materials.

Experiment 1a: Hydrogen Storage Capacity of Hydrochloric Acid via Reaction with Zinc

Granular zinc metal (2 g) is placed in a 100 mL round-bottom flask with a glass stopcock side arm port (Kemtech America, Inc.). A plastic adapter is connected with plastic tubing from the stopcock on the round-bottom flask to the fuel cell (Fuel Cell Store, Dr. Fuel Cell Dismantlable Fuel Cell Extension Kit), which has a measurement box (Dr. Fuel Cell Load Measurement Box) that reports voltage and current during movement of a motor on the box. Another plastic three-way adapter is connected to a pressure/temperature sensor with tubing to both the fuel cell and an ideal gas syringe (PASPORT Ideal Gas Law Apparatus, TD-8596A)¹⁴ as shown in Figure 1. To determine the hydrogen storage capacity of hydrochloric acid via reaction with zinc, the stopcock/pinch clamp connected to the ideal gas law apparatus is closed, and a septum stopper is attached to the 24/40 joint on the flask in order to add 8.0–10.0 mL of 3.0 M hydrochloric acid all at once using a syringe to produce gaseous hydrogen. Figure 2 shows the equipment setup in a vacuum hood. Gaseous hydrogen is formed as a product of the single replacement in reaction 4.



The energy carrier, hydrogen, is then converted into electric energy by the fuel cell, and readings are taken of the current, voltage, and time when the motor is running.

Table 1. Measurements for the Reaction of 2.0 g of Zn with 8.8 mL of 3.0 M HCl Solution

<i>t</i> (s)	<i>I</i> (amps)	<i>V</i> (volts)	<i>W</i> (watts)	H ₂ (g(10 ⁻³)) ^a	H ₂ (mol(10 ⁻³)) ^a
10	0.014	0.74	0.010	0.15	0.074
30	0.014	0.74	0.010	0.44	0.22
60	0.014	0.73	0.010	0.88	0.44
90	0.014	0.73	0.010	1.3	0.63
150	0.014	0.72	0.010	2.2	1.1
210	0.014	0.71	0.0099	3.1	1.5
270	0.014	0.71	0.0099	4.0	2.0
330	0.014	0.70	0.0099	4.8	2.4
390	0.014	0.70	0.0098	5.7	2.8
450	0.014	0.70	0.0098	6.6	3.3
510	0.014	0.70	0.0098	7.5	3.7
570	0.014	0.70	0.0098	8.3	4.1
630	0.014	0.70	0.0098	9.2	4.6

^aConsumption of hydrogen in grams and moles.

Results and Discussion

Experiment 1a: Hydrogen Storage Capacity of Hydrochloric Acid via Reaction with Zinc

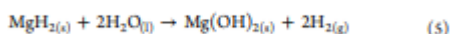
Table 1 was created from voltage and current readings of the fuel cell when hydrochloric acid was reacting with granular Zn. The data were collected during the first 10.5 min of the reaction, which was far from completion. When the data for the time of the running motor, *t* in seconds, voltage, and current were collected, the moles of hydrogen consumed by the fuel cell were calculated for each step using eq 7, where *n* is the number of moles of electrons transferred (two per mole of hydrogen), and *F* is Faraday's constant (96,485 C/mol)

$$\text{Mass of hydrogen gas consumed} = (I)(t)(\text{Molar Mass}_{\text{H}_2})/(nF)$$

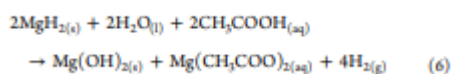
The voltage and current at each measurement time consistently were about 0.7 V and 0.014 A, respectively, for a power of 0.01 W. Power was calculated from the product of voltage *V* in volts and current *I* in amps. The reaction of 2.0 g of Zn with 8.8 mL of 3.0 M HCl, where HCl is the limiting reagent, would theoretically produce 0.027 g of gaseous hydrogen. The hydrogen consumption in the fuel cell by the time the measurement was stopped was only 0.34% of the theoretical yield. The calculation based on eq 7 shows that 100% yield of gaseous hydrogen, when the reaction goes to completion, would provide electrical energy to run the 0.01 W motor for about 50 hours.

Experiment 1b: Hydrogen Storage Capacity of Magnesium Hydride

Hydrogen production during hydrolysis of MgH₂ in the presence of acetic acid is measured. As shown in reaction 5, gaseous H₂ is released when solid MgH₂ reacts with liquid H₂O.



However, the solid Mg(OH)₂ product on the MgH₂ surface prevents further reaction. Therefore, acetic acid is added to make soluble magnesium acetate (reaction 6), which allows reaction to occur with the inner layers as the reaction front moves deeper into the MgH₂ solid.^{15,16}



MgH₂ in the presence of acetic acid, initially, the pinch clamp/ stopcock is shut off to the fuel cell so that the hydrogen produced is directed toward the sensor. The MgH₂ (VWR International) sample with the mass 100–150 mg (the MgH₂ is a limiting reagent and water is in excess) is placed inside

the dry flask and capped with a septum stopper. The 5.0 mL of 2.0 wt % acetic acid is added using a syringe and needle through the septum stopper to ensure that the hydrogen produced in reaction 6 is not released to the surroundings. The volume of the gas law syringe is kept constant during the hydrolysis. The pressure/temperature sensor plugs into a PASPORT USB Link, PS-2100A, which connects to a computer's USB port for data collection. The DataStudio software¹⁷ is fully compatible with the PASPORT sensor, and displays may include digits, graphs, meters, and tables. For example, a plot of pressure versus time data may be displayed on the computer. Fuel cell consumption of hydrogen during hydrolysis of MgH₂ in the presence of acetic acid is measured. After the hydrolysis, the pinch clamp/stopcock to the fuel cell is opened to allow the hydrogen to reach the fuel cell, and measurements are recorded of current, voltage, and time the motor is running. Hydrogen storage capacity of MgH₂ is determined by thermogravimetric analysis (TGA). TGA of the MgH₂ sample is provided to the students by the instructor to determine the weight loss when hydrogen is released with an increase in temperature. TGA was conducted in an atmosphere of gaseous nitrogen. A sample of the magnesium hydride with the mass 25.617 mg was heated in a platinum crucible with a temperature rate of 20 °C/min up to 500 °C and held at isothermal conditions for 2 min. The weight loss analysis showed that the sample lost 3.50 mass % by 375 °C. The weight loss corresponds to the actual hydrogen capacity of the sample used in this study.

Results and Discussion

Experiment 1b: Hydrogen Storage Capacity of Magnesium Hydride The reaction of 149.7 mg of MgH₂ with 5.0 mL of aqueous 2.0 wt % acetic acid results in a nearly linear increase in the pressure as a function of reaction time when the measurement was discontinued at 90 min, far from completion (Figure 3). The average rate of hydrolysis, which was calculated using the ideal gas law to convert pressure to moles, was found to be $(3.7 \pm 0.2) \times 10^{-7} \text{ mol H}_2 \text{ s}^{-1}$. Calculations show that after 90 min duration of the reaction, the mass loss of the MgH₂ sample is 2.3%. To calculate moles of hydrogen released and convert them to grams, the volume of the system was needed. Assuming ideal gas behavior for a fixed amount of hydrogen, the volume of the system in Figure 2, V_o, was calculated using eq 8 when the stopcock is turned off toward the fuel cell, and the volume of the syringe was changed from the initial, V_i, to final volume, V_f.

$$(V_i + V_o)/(V_f + V_o) = (T_f P_i)/(T_i P_f) \quad (8)$$

The volume of the system was found to be $110 \pm 8 \text{ mL}$. Similar to the experiment with HCl, data were collected for the time of the running motor, voltage, and current, and using eq 7, the amount of hydrogen consumed by the fuel cell was calculated. The voltage, current, and power results for the operation of the fuel cell during the MgH₂ experiment were within experimental error of the data reported for the HCl reaction with Zn in Table 1. The hydrolysis of 100.7 and 147.3 mg of MgH₂ with 5.0 mL of 2.0 wt % acetic acid consistently showed a rate of hydrogen consumption of $6.7 \times 10^{-8} \text{ mol s}^{-1}$; the mass of hydrogen consumed by the fuel cell was 0.13% after 78.2 and 47.0 min of reaction time for two samples, respectively. The actual hydrogen storage capacity of the MgH₂ sample in this experiment was determined by TGA weight loss measurements to be 3.5%. For the fuel cell consumption experiments, the measured weight loss of 0.13% is only 4% on the TGA value, primarily because the reaction was stopped far away from completion, and additionally, a large amount the hydrogen, which is produced, remains in the volume outside of the fuel cell. Calculations show that the actual hydrogen capacity of the MgH₂ samples should be enough for more than 35 h of electricity generation by the fuel cell. The Supporting Information is provided for the experiments and shows

additional figures, sample calculations, and the TGA weight loss for MgH_2 as a function of temperature.

Hazard

Hydrogen is the lightest of the elements and therefore dissipates rapidly in open areas and migrates through very small spaces. It normally is present only in the gaseous state since its boiling point is $-252\text{ }^\circ\text{C}$. Hydrogen should be stored safely and effectively. The immediate dangers are of fire or explosion in enclosed environments. Sensors often are installed to detect concentrations of hydrogen approaching 4%; the lowest percentage in the atmosphere at which ignition can occur, referred to as the 100% lower explosion limit (LEL). The septum stopper should not be wired onto the 24/40 joint of the flask since it serves as a potential pressure vent in case of the buildup of high pressure. Granular Zn should be used, never powdered Zn. MgH_2 should be stored in a desiccator to avoid contact with water, and a dry flask should be used for the reaction. Take care when using the syringe since accidental needlesticks may cause infection

Conclusion

Hydrogen storage capacity of hydrogen-containing compound experiments were developed for use in workforce training associated with clean energy technologies. Undergraduate students gained valuable experience in (1) examining the production of hydrogen stored in aqueous hydrochloric acid and solid magnesium hydride, (2) consuming hydrogen with a fuel cell, (3) analyzing and comparing the storage capacity of two hydrogen-containing compounds, and (4) making measurements using the ideal gas equation of state.

Topic 4: Hydrogen Fuel-Cell

4.1 Hydrogen Fuel Cell

The popularity of hydrogen fuel is growing by the day, owing to the fact that when burned, it creates a tremendous amount of heat.

For most of us, the fact that hydrogen fuel cells can discharge more energy than diesel or petrol may come as a shock. This indicates that a hydrogen fuel cell is roughly three times as efficient as a gasoline engine. In addition, when compared to the usage of gasoline, this fuel produces fewer pollutants.

Although hydrogen is the most common fuel, fuel cells also require oxygen. One of the most appealing features of fuel cells is that they produce power with very little pollution; much of the hydrogen and oxygen needed to create electricity eventually mix to form a benign by-product, water. An electrolyte transports electrically charged particles from one electrode to the other, while a catalyst speeds up electrode reactions in each fuel cell.¹⁰

4.1.1 What is a Fuel-Cell?

A fuel cell is an electrochemical cell that uses an electrochemical reaction to create electrical energy from fuel. To keep the processes that generate electricity going, these cells need a constant supply of fuel and an oxidizing agent (usually oxygen). As a result, until the supply of fuel and oxygen is shut off, these cells can continue to generate power. Fuel cells, despite being conceived in 1838, did not enter commercial usage until a century later, when NASA utilized them to power space capsules and satellites. Many establishments, including businesses, commercial buildings, and residential buildings, now use these devices as a major or secondary source of electricity. A cathode, anode, and electrolyte make up a fuel cell, which is comparable to an electrochemical cell. The electrolyte in these cells allows protons to travel around. Fuel cells, like batteries, produce electricity without causing combustion or pollution. Fuel cells, unlike batteries, do not run out of power or need to be recharged as long as there is a continual supply of fuel and oxygen. When compared to traditional gasoline vehicles, fuel cell vehicles can reduce carbon dioxide emissions by up to half when hydrogen is created from natural gas and up to 90% when hydrogen is produced from renewable energy sources like wind and solar. In addition, the tailpipe emits no pollutants—only water!¹¹

¹⁰ [What is Hydrogen Fuel Cell - Definition, Production, Advantages, Economy & Uses. with Videos \(byjus.com\)](https://byjus.com/chemistry/hydrogen-fuel-cell/)

¹¹ [5 Fast Facts about Hydrogen and Fuel Cells | Department of Energy](https://www.energy.gov/eere/vehicles/articles/5-fast-facts-about-hydrogen-and-fuel-cells)

4.1.2 Working Principle of a Fuel Cell

A fuel cell can use the chemistry between hydrogen and oxygen to generate power. This type of cell was utilized in the Apollo space program and had two purposes: as a source of fuel and as a source of drinking water (the water vapour produced from the cell, when condensed, was fit for human consumption).

This fuel cell worked by transferring hydrogen and oxygen through carbon electrodes into a concentrated sodium hydroxide solution. The following is a formula for the cell reaction:

Cathode Reaction: $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

Anode Reaction: $2\text{H}_2 + 4\text{OH}^- \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$

Net Cell Reaction: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

This electrochemical process, however, has a slow reaction rate. A catalyst, such as platinum or palladium, is used to solve this problem. Before being inserted into the electrodes, the catalyst is finely split to maximize the effective surface area. This fuel cell's block diagram is shown below.

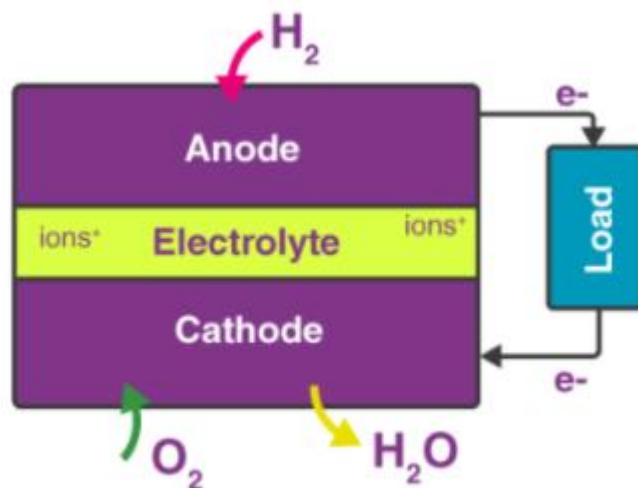


Figure SC-E-SRB- 7: Sample Hydrogen Fuel Cell

The above-mentioned fuel cell has a 70 percent efficiency in the generation of electricity, whereas thermal power plants have a 40 percent efficiency. Because the creation of electric current in a thermal power plant requires the conversion of water into steam and the use of that steam to move a turbine, there is a significant variation in efficiency. Fuel cells, on the other hand, provide a platform for converting chemical energy into electrical energy directly.

4.2 Types of Fuel Cells

Fuel cells come in a variety of shapes and sizes, despite the fact that they all work in the same way. In this part, we'll look at a few of these fuel cell kinds.

The Polymer Electrolyte Membrane (PEM) Fuel Cell

- Proton exchange membrane fuel cells are another name for these cells (or PEMFCs).
- These cells work in a temperature range of 50 to 100 degrees Celsius.
- A polymer with the ability to conduct protons is employed as the electrolyte in PEMFCs.

- Bipolar plates, a catalyst, electrodes, and the polymer membrane make up a conventional PEM fuel cell.
- Despite their environmentally favorable applications in transportation, PEMFCs can also be utilized for stationary and portable power generation.

Phosphoric Acid Fuel Cell

- In order to channel the H^+ , these fuel cells use phosphoric acid as an electrolyte.
- The working temperatures of these cells are between 150 and 200 degrees Celsius.
- Because phosphoric acid is non-conductive, electrons are forced to travel to the cathode via an external circuit.
- The electrolyte's acidic nature causes the components of these cells to corrode or oxidize over time.

Solid Acid Fuel Cell

- The electrolyte in these fuel cells is a solid acid substance.
- At low temperatures, the molecular structures of these solid acids are organized.
- A phase transition can occur at higher temperatures, resulting in a massive increase in conductivity.
- $CsHSO_4$ and CsH_2PO_4 are examples of solid acids (cesium hydrogen sulfate and cesium dihydrogen phosphate respectively)

Alkaline Fuel Cell

- In the Apollo space program, this was the fuel cell that was employed as the major source of electricity.
- An aqueous alkaline solution is utilized to saturate a porous matrix in these cells, which then separates the electrodes.
- These cells' working temperatures are extremely low (approximately 90°C).
- These cells are quite effective. Along with electricity, they also produce heat and water.

Solid Oxide Fuel Cell

- A solid oxide or ceramic electrolyte is used in these cells (such as yttria-stabilized zirconia).
- These fuel cells are both very efficient and inexpensive (theoretical efficiency can even approach 85 percent).
- These cells operate at extremely high temperatures (lower limit of 600°C, standard operating temperatures lie between 800 and 1000°C).
- Because of their high operating temperatures, solid oxide fuel cells are only suitable for fixed applications.

Molten Carbonate Fuel Cell

- Lithium potassium carbonate salt is employed as an electrolyte in these cells. At high temperatures, this salt turns liquid, allowing carbonate ions to move around.
- These fuel cells, like SOFCs, have a relatively high working temperature of 650°C.
- Because of the high working temperature and the presence of the carbonate electrolyte, the anode and cathode of this cell are prone to corrosion.
- Carbon-based fuels such as natural gas and biogas can be used to power these cells.

Because hydrogen employs nitrogen as an impurity, the fuel contains nitrogen oxides as contaminants (here we are talking about the molecule of hydrogen and not the atom). If a little water is put to the container to lower the temperature, pollution can be reduced. The oxygen-nitrogen reaction would be stifled as a result. A hydrogen gas cylinder in compressed state would be much heavier than a petrol tank containing the fuel. The temperature of the hydrogen gas must be pushed down to 20 K in order to turn it into a liquid. It is difficult to use this fuel efficiently due to these limits. Hence, experts around the world are conducting researchers to find ways on how to use this fuel in a simple way in the future.

4.3 Components of Hydrogen Fuel Cells

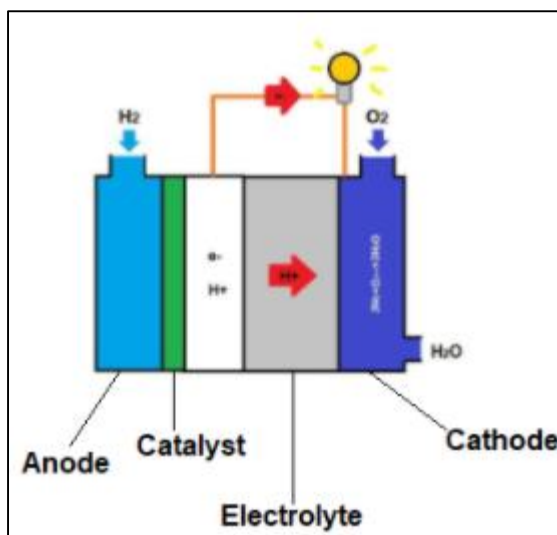


Figure SC-E-SRB- 8: Components of a Hydrogen Fuel Cell (Source: [Fuel Cell - Definition, Working, Types, and Applications of fuel cell. \(byjus.com\)](#))

4.3.1 Hydrogen Fuel Cell Working Principle

Hydrogen and fuel cells (FCs) are often regarded inseparable since hydrogen-fueled FCs have the best performance and have the smallest environmental impact of all the FCs. Hydrogen, like electricity, is a carrier of energy rather than a fundamental source of energy. There are no pure hydrogen reserves in nature. Water can be electrolyzed to produce hydrogen in the opposite direction of the fuel cell reaction. Significant amounts of electricity, much of which is now generated by the burning of fossil fuels, are required to cause electrolysis to occur.

However, not all hydrogen fuel cells are designed to be used in automobiles. Stationary fuel cells for power generation have been in the works for decades. The creation of electricity from fuel inputs such as natural gas or hydrogen and the utilization of waste heat from that process to heat the building could be some of the uses of fuel cells to residential or commercial structures.

At the anode of a fuel cell, hydrogen atoms are stripped of their electrons through a chemical reaction. Ionized hydrogen atoms now have a positive electrical charge. Electrons with a negative charge provide the current that moves across wires. The DC output of the fuel cell must be channeled via a conversion device called an inverter if alternating current (AC) is desired.

The fuel cell gets its name from the hydrogen supply, which is frequently referred to as the fuel, despite the fact that there is no combustion involved. Then, electrochemically, hydrogen oxidation

takes place in a very efficient manner. During oxidation, hydrogen atoms react with oxygen atoms to generate water; electrons are liberated and travel as electric current through an external circuit.

A fuel cell is an electrochemical system that generates energy by combining hydrogen and oxygen, with water and heat as byproducts. In its most basic form, a single fuel cell comprises of two electrodes—an anode and a cathode—with an electrolyte in between. At the anode, hydrogen reacts with a catalyst to produce a positively charged ion and a negatively charged electron.

4.3.2 Advantages of Hydrogen Fuel

- On burning hydrogen, it emits only water vapour,
- When hydrogen is burned it does not produce carbon dioxide.
- Hydrogen deletes little tailpipe pollution and is considered less of a pollutant.
- It has the ability to run a fuel-cell engine when compared to an internal combustion engine.

4.4 Application of Hydrogen Fuel in Vehicles

Vehicles today use a number of fuels, some of which you may not be aware of. While the vast majority of vehicles run on gasoline, there is technology to burn different fuels to satisfy our transportation demands, each with its own set of benefits and drawbacks.

- **Gasoline** – Today, gasoline, often known as petrol, is the most commonly utilized fuel in automobiles. This particular fossil fuel is made for four-stroke engines, such as those found in most automobiles. According to the University of Michigan's fossil fuels website, gasoline offers for quick starting, acceleration, easy combustion, and quiet operation. The hydrocarbons in gasoline, as well as the carbon dioxide produced when it is burned, contribute to pollution, smog, and global warming. Despite being the most readily available fuel, it is seen as a transient source of energy due to its high cost, negative environmental effects, and scarcity of resources.
- **Diesel** – Transport vehicles such as tractor-trailer trucks, buses, boats, and railroads all utilize diesel fuel. Like gasoline, this fossil fuel is non-renewable. Diesel emits more chemical compounds and nitrous oxide, which contribute to pollution, although emitting less carbon dioxide. According to the Petrol Prices website, diesel automobiles last longer than gasoline vehicles and have 30 percent greater fuel efficiency than the average gasoline vehicle.
- **Liquefied Petroleum** – Propane, or liquefied petroleum, is a clean fuel alternative to gasoline that is utilized in a limited number of automobiles. When liquefied petroleum is burned, it creates fewer toxins and does not contribute to smog like diesel and gasoline do. Propane also has a lower cost than gasoline.
- **Compressed Natural Gas** – Compressed natural gas, or CNG, can be used in gas and diesel engines. CNG is a transparent, odorless, and non-corrosive gas that can be utilized to power a combustion engine as a liquid or a gas. According to the Consumer Energy Center website, vehicles equipped with a CNG fuel system emit 80% less ozone-forming emissions than

gasoline-powered vehicles. CNG filling stations can be found in a few places across the US, mostly in California.

- **Ethanol** – Ethanol is a biofuel that is produced by converting sugar cane, corn, barley, and other natural resources into gasoline. Ethanol has grown in popularity as a fuel since it is one of the few fuels that can be used in a gasoline engine without any modifications. Although several car types can operate entirely on ethanol, it is more usually used as an addition. To help reduce emissions and contamination caused by pure gasoline components, some jurisdictions have enforced the use of ethanol. E10, or gasoline containing 10% ethanol, is now available at most gas stations in the United States. In some regions, even higher concentrations are used.
- **Bio-diesel** - Sugar beet, rapeseed, or palm oil are used to make bio-diesel, which is a diesel replacement. This chemical is sometimes made by collecting leftover oil from restaurant fryers. When compared to regular gas or diesel, bio-diesel burns cleaner and emits significantly less CO₂. Continuing to produce this material, however, may result in excessive deforestation..¹²

Now, we have listed all these fuels that vehicles run on. Most vehicles in Papua New Guinea and in the world, run on fuel from non-renewable sources such as Gasoline or Petrol and Diesel. Scientist and Researchers have looked into the potential of using Hydrogen and Fuel Cells to power up Vehicles, as one application of it.

In this topic, we will be comparing the potential of Hydrogen fuel powering vehicles against the normal Gasoline and Diesel fuels.

4.5 The Challenges of The Hydrogen Electric Car

Let's start with some background information. The term "hydrogen electric car" refers to a vehicle that runs on hydrogen and is powered by a particular device called a fuel cell. The hydrogen fuel cell vehicle is one of the large family of electric vehicles, as it relies on an electric engine for propulsion. Electrical energy is simply stored in the battery after charging on the electrical grid for an electric car powered by a lithium-ion battery. Herein lies the distinction between a hydrogen car and a conventional automobile. The powertrain's electricity is provided not just by the battery, but also by the fuel cell, which uses hydrogen stored aboard the car. The majority of hydrogen vehicles currently produced by automobile manufacturers work in this way.

The hydrogen-powered car aspires to achieve the same emission-free driving goals as the "traditional" electric vehicle. Nonetheless, various obstacles must be solved in order to limit the environmental impact of hydrogen production to the greatest extent possible. Because it is extracted from fossil fuels, the present method of harvesting this chemical element relies on steam reforming of hydrocarbons. It is referred to as "grey" hydrogen.

However, there is another method for obtaining hydrogen: water electrolysis. It is feasible to manufacture "green" hydrogen when the electricity utilized in the electrolysis process comes from a renewable energy source such as solar or wind power. This extraction method is a solution for the future.

¹² [Six Types Of Fuels Used In Today's Vehicles - MyMoto Nigeria](#)

4.6 How The Hydrogen Electric Vehicle Works Using A Fuel Cell

In actuality, how does a hydrogen fuel cell vehicle work? A fuel cell provides the electric energy. Onboard the vehicle, pressurized hydrogen is stored in special tanks. The fuel cell is supplied with hydrogen (H_2) and dioxygen (O_2) from the surrounding air. These two gases then perform an electrochemical reaction inside the cell, resulting in the production of energy, heat, and water vapor (H_2O), which is expelled as a gas through a tiny tube beneath the car.

The generated energy, along with energy from the battery, is used to power the car's electric motor, which may then operate silently and emit no pollutants or CO_2 emissions*. When it comes to refueling, hydrogen is replenished at designated stations utilizing pumps that feed hydrogen into the vehicle's tank in the form of compressed gas very quickly.

4.7 The Energy Consumption of a Hydrogen Fuel Cell Vehicle

What is the fuel consumption of a hydrogen fuel cell electric vehicle? Keep in mind that the hydrogen atom is one of the simplest and lightest natural elements on the periodic table, requiring two to form the dihydrogen (H_2) molecule (the classification which lists all the chemical elements present on Earth.) It has an extremely low density of volume. As a result, obtaining the amount of hydrogen required to power a vehicle necessitates storing a significant amount in massive tanks under high pressure.

4.8 The Hydrogen Fuel Cell Electric Vehicle According to Renault Group

A hydrogen fuel cell electric vehicle is an electric vehicle that combines a lithium-ion battery with a hydrogen fuel cell under one hood, according to Renault Group. By combining the two technologies, the brand achieves the best of both worlds. Renault has incorporated a fuel cell behind the renowned diamond emblem on its electric utility car Kangoo Z.E. Hydrogen.

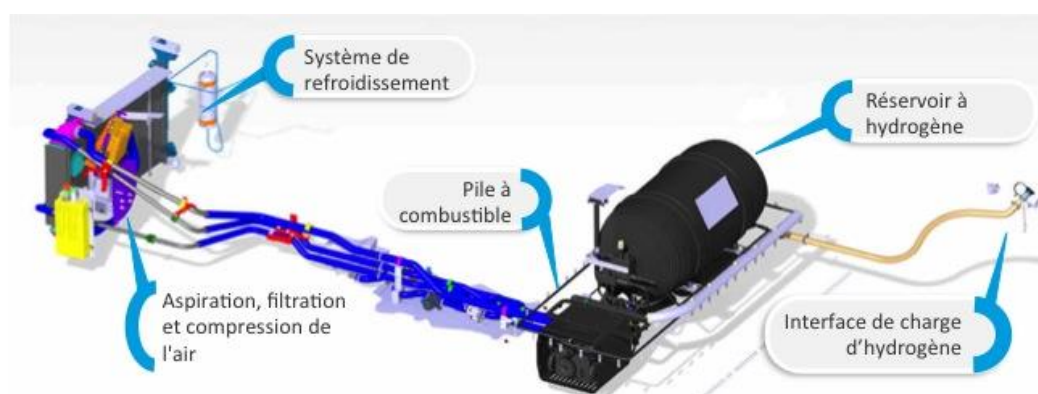


Figure SC-SRB-2: Diagram of the fuel cell installation onboard the Renault Kangoo Z.E. Hydrogen.

The motor of the hydrogen-fueled electric automobile is powered by its lithium-ion battery in this Kangoo Z.E. Hydrogen operational diagram. The fuel cell, on the other hand, adds to the car's range by providing an additional energy reserve (providing more autonomy and shorter charging time.) Hydrogen-powered vehicles provide the driver the practical and budgetary benefits of electric power, as well as the flexibility and range benefits of hydrogen.

Sample Module Activities

Activity 1: Group Work/Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research on their understanding of how a Hydrogen Fuel-Cell works. Their research should cover innovative applications of a Hydrogen fuel cell. Students should also come up with their own innovative applications of Hydrogen Fuel Cell to improve modern technology in context.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 2: Practical Assignment or Practical Experiment

Title: Demonstrating the Principle of how Fuel Cells Work

Source Link: https://www.scienceinschool.org/article/2014/eco_energy/

To explore how fuel cells work, we have developed a low-cost fuel cell for use in the classroom. The resulting electrolyzer and fuel cell can be used as a part of a setup to demonstrate how hydrogen might be produced and used.

The following criteria for the models were chosen:

1. Materials must be accessible and cheap
2. The whole construction should be comparatively easy to allow others to observe the phenomena and processes
3. The prototypes should be safe to make

Here we describe the materials and procedures required for constructing a water electrolyzer and hydrogen fuel-cell prototype for the classroom.

Materials and Reagents

- Plastic Syringes (50ml)
- Graphite Rods or Pencil Leads
- Carbon Gas-Diffusion Electrodes pressed on Ni-mesh
- Small flexible Piping with valves
- T-shaped glass connector
- Plastic Vessel (bottle) for the electrolyte
- Laboratory stands with clamps
- Aluminum or copper foil
- Grease-proof paper
- Hot-plate
- Glass vessels (400ml beakers)
- Isolated copper wires with connectors
- Drill
- Silicon Paste
- 6-12V DC power source (Solar Panel or 9V battery)
- End Consumer (low-powered electromotor with fan, LED)
- Set of Resistors

- Two multimeters, or a voltmeter and ammeter
- 1M H_2SO_4
- 2M HNO_3 (for etching)
- PdCl_2 solution prepared by dissolving 1g of the salt in 50mL of 0.5M HCl . Heat to 50°C and then dilute to 100ml with distilled water.
- 1M KOH or NaOH
- Protective Gear: Gloves, Goggles, Shoes, and Coat

Procedure

The steps marked with asterisks (*) should be performed by the teacher for safety.

Constructing the Electrolyzer

1. * Remove the plungers and needles from two syringes
2. * Remove the rubber gaskets from the plungers and make a hole for a graphite rod electrode in each of the rubber gaskets.
3. Push the graphite rods into these holes and insert the rubber gaskets back into the syringes.
4. * Drill holes into the sides of both syringes about 1 cm from the bottom. The holes should be about 1mm bigger than the outer diameter of your T-shape connector.
5. Attach 1cm lengths of hose to both edges of the T-shape connector.
6. Connect the syringes with the T-shape connector
7. Use the hose to connect the electrolyte container with the free edge of the T-shape connector.
8. Connect the thin flexible tubes to the narrow ends of the syringes.
9. * Pour 1M H_2SO_4 into the electrolyte container until the electrolyte fills both syringes
10. Connect both electrodes by copper wires to the power source (Figure 1)

Constructing the Fuel Cell

1. * Remove the plungers and needles of two syringes
2. * Cut two pieces from gas-diffusion electrodes to completely cover the flange of the syringes.
3. * Place a piece of aluminum or copper foil on a hot-plate, put grease-proof paper on it, and then place the shaped gas-diffusion electrode on the paper.
4. * Press the flange of a syringe onto the gas-diffusion electrode – due to the high temperature, the plastic will melt and the electrode will stick to the syringe.
5. * Drill a hole through one side of the flange and the sealed gas-diffusion electrode.
6. Push a metal screw through the hole
7. Remove the insulation from one end of a piece of copper wire, make several loops around the screw and tighten it with a nut.
8. Isolate the bolt with silicon paste.
9. Repeat steps 3 to 8 to make the second electrode.
10. * Etch the electrodes by immersing them in 2M HNO_3 for 5 minutes, then wash with water.
11. * Pour the solutions of PdCl_2 into the 400ml beaker and immerse the prepared electrodes in it for 5 minutes. Based on the reactivity series of metals, a thin catalytic layer of elemental palladium is deposited on gas-diffusion electrodes when palladium ions come into contact with the carbon and nickel mesh.
12. * Pour 100mL 1M NaOH (or KOH) into another beaker and place both syringes with sealed electrodes in it (Figure 2)
13. Connect these syringes to the two thin hoses from the electrolyzer

After collecting some quantity of hydrogen and oxygen, stop the gas production and open the valves to allow the gases to pass to the electrodes of the fuel cell (Figure 4). Students can then measure the electrical parameters of the fuel cell by using the circuit described in Figure 5. The current, I , is calculated according to Ohm's law:

$$I = V/R$$

Instead of a resistor, a light emitting diode (LED) or low-power electromotor can also be used.

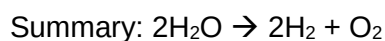
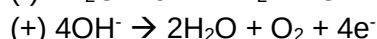
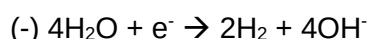
An important advantage of this system is that each of the basic modules can be replaced by other devices. For example, a special wind turbine can be used instead of a solar panel to generate the electricity that is necessary to supply the water electrolyzer. Or, instead of a water electrolyzer as a source of gaseous hydrogen and oxygen, gas generation by chemical reaction (Figure 6) can be used.

Other version of a fuel cell, using liquid fuel (for example, ethanol instead of hydrogen), can also be developed. With some basic theoretical background knowledge, students can use this skill to develop prototypes of advanced and sophisticated technologies (Figure 7).

Practical Demonstration

Linking together the electrolyzer and fuel cell with a solar panel, as shown in Figure 1, can demonstrate how solar energy can be stored as hydrogen and then converted back into electricity. The electricity needed to power the water electrolysis can be generated by shining an artificial light source or the sun onto the solar panel, after which the evolved gases are collected above the electrolyte in the separated parts of the electrolyzers (syringes). Valves stop the gases passing from the electrolyzer to the fuel cells until it is needed.

The syringes also help to show the volume of the gas from the anode is twice the volume of the gas from the cathode: 2 moles of hydrogen and 1 mole of oxygen are produced from 2 moles of water (Figure 3).



Activity 2: Exercises

Complete the following questions.

1. What is hydrogen and fuel cells?
2. How efficient is a hydrogen fuel cell?
3. How much power does a hydrogen fuel cell produce?
4. What are the advantages of a hydrogen fuel cell?
5. What type of reaction is a hydrogen fuel cell?
6. What is a disadvantage of hydrogen fuel cells?

THEMA: BATTERY TECHNOLOGY

Introduction

A battery is a storage device and converter for chemical energy and electrical energy. It has one or more electrochemical cells that are connected to external circuits. The resulting conversion of the stored chemical energy to electrical energy powers the electrical devices such as flashlights, electric cars and mobile phones. When looking at the battery, the positive terminal is known as the cathode while the negative terminal is known anode.

Electrons flow from the anode then to the external circuit and onto the cathode. When the battery is connected in a circuit to give electricity, there is a reaction where electrons are gained and lost and this reaction is known as the reduction-oxidation reaction or in short, the redox reaction. The redox reaction in the battery converts the stored high chemical energy which reacts to low energy products and the free energy difference between the high energy of the reactants and the low energy of the products results as electrical energy to the external circuit.

Listed below are few of the many career pathways for students who are interested in the field of battery technology development.

- Cell Developer
- Process Engineering Technician
- Mechanical Engineer (Battery Performance Testing)
- Battery Product Design Engineer – Cell Mechanical Characterization
- Electrode Coating Process Chemistry/Engineer
- Battery Cell Engineer, and etc.

MODULE 12.1: SOLID-STATE BATTERY

A lot of technology we use today was not around a while back such as smartphones which are thousand times faster than the computer that powered the first moon landing.

From iPods to Laptops and smart phones, the lithium-ion battery is at center of so much of our modern-day technology. Since they were first being research in the 1970s and 1980s to eventually becoming commercially produced by SONY in 1991. They have improved in terms of:

- Energy Density
- How fast they can charge and discharge
- How Safe they are
- Affordability

Once Tesla started to manufacture electric vehicles and built the Giga factories, the level of Lithium-ion production around the world has risen by such a factor that prices lowered to a point where a 480km mile range electric vehicles are completely viable. But as great as today's batteries are, they are starting to hit some limitations and are not going to be able to get better in terms of energy density and life cycles without a huge breakthrough. That breakthrough is the **Solid-State Battery**.

Battery technology could be a game-changer in the energy transition, allowing for the decarbonization of the transportation sector while also providing a necessary backstop for power generation from intermittent solar and wind sources. However, the widely used lithium-ion battery may not be up to the task of powering the global green economy's future.¹³ Solid-state batteries have been touted as a long-term replacement for Li-ion batteries. Although research is continuing and prototypes are being developed, it could be a decade before a solid-state device is commercially available.

¹³

<https://www.forbes.com/sites/arielcohen/2021/02/11/what-batteries-will-power-the-future/?sh=5ce1b64841c0>

Topic 1: Introduction to Solid-State Battery

Key Concepts:

- Lithium-ion battery vs. Solid-State Battery
- Reasons to develop Solid-State Battery
- How a Solid-State Battery Works
- Pros and Cons
- Challenges

1.1 Li-ion Battery vs Solid-State Battery

To understand why the Solid-State Battery is going to be one of the most innovative breakthroughs of the 21st Century, we have to talk about batteries in general. Let us use the Lithium-ion battery as an example.

1.1.1 Components of General Battery

Most batteries have the following main components:

- Cathode: The Positive (+) Electrode, meaning it is electrically conductive
- Anode: The Negative (-) Electrode, also meaning it is electrically conductive
- Separator: An Electrical Insulator that prevents short circuits and is usually a porous material
- Electrolyte: *a conductive chemical mixture that allows the flow of electrical current between the two electrodes*. In this case, it is a liquid solution through which lithium ions move from the anode to the cathode

1.1.2 Lithium-ion (Li-ion) Battery

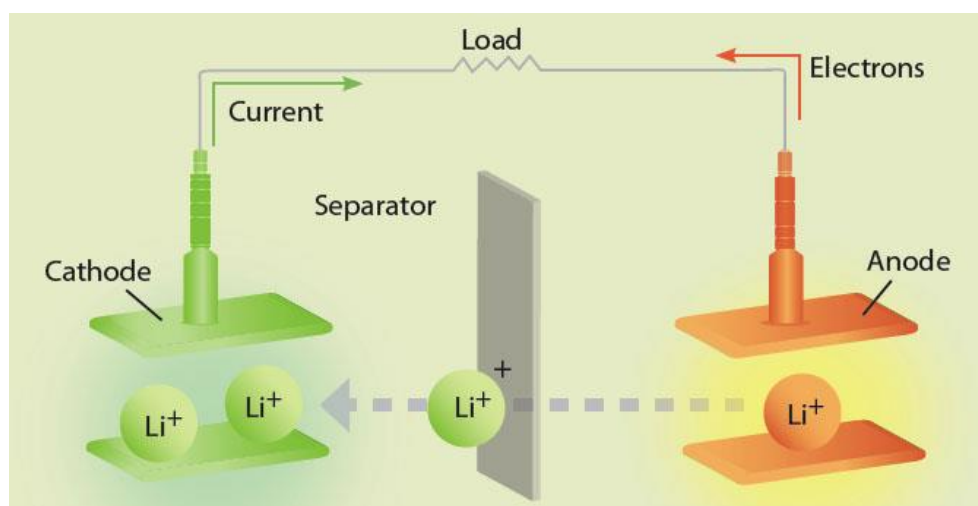


Figure SC-BAT-SRB- 1: How a Lithium-Ion Battery Works

Lithium-ion batteries function by allowing ionic lithium to pass through the electrolyte barrier between the anode and cathode of the battery (the positive and negative ends). Positively charged lithium ions travel through a separator in a conducting liquid electrolyte to reach the cathode in ordinary Li-ion batteries. The anode now has a negative charge of electrons. This flow is reversible when charging.

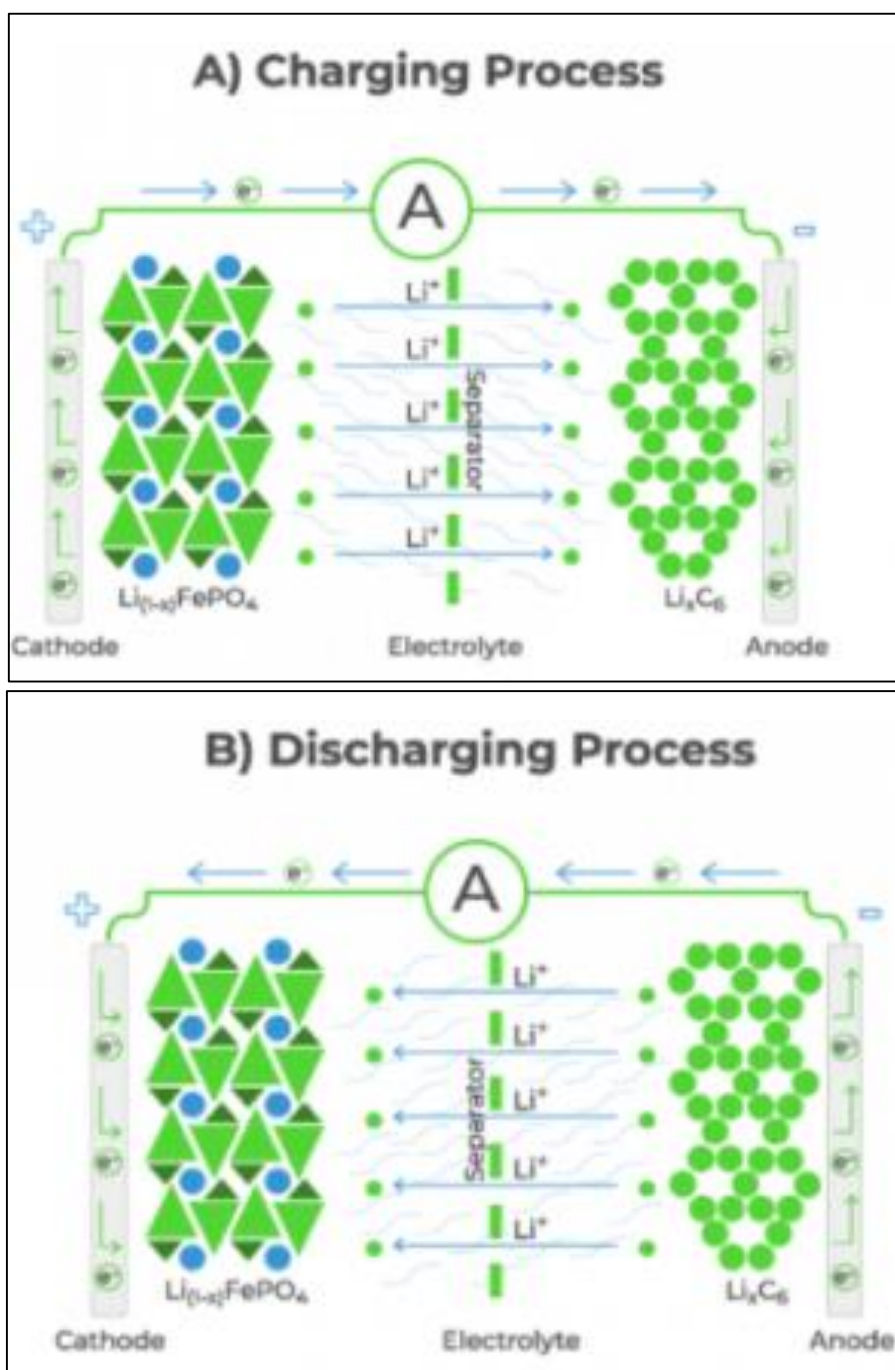


Figure SC-BAT-SRB- 2: Lithium-Ion Battery

When a device, such as a light bulb, is connected to a battery, a chemical reaction takes place between the anode, cathode, and electrolyte, resulting in a flow of electrical energy to power the light bulb.

Solid-State Battery

A Solid-State Battery, as the name suggests, is a type of battery that uses a solid electrolyte rather than the liquid or polymer gel electrolytes used in lithium-ion or lithium-polymer batteries. The electrolyte is the major distinction between a lithium-ion battery and a solid-state battery. Solid-state batteries employ a solid electrolyte, whereas lithium-ion batteries (and most other batteries) use a liquid electrolyte. Continue reading to learn more about these issues and what they entail for battery users.¹⁴



Figure SC-BAT-SRB- 3: The charge and discharge process of a LiFePo₄ battery works by moving Lithium Ions across the electrolyte while electrons pass through the circuit.

A lithium-ion battery uses a liquid electrolyte to regulate the flow of current, and a solid-state battery uses a solid electrolyte.¹⁵

1.2 Reasons to develop Solid-State Battery

Solid-state battery technology has numerous advantages. In a nutshell, a solid-state battery's solid electrolyte provides improved energy density, a longer lifespan, and increased safety in a smaller package!

Let's take a closer look at how this translates to how solid-state battery technology will benefit a user:

1. **Smaller and More Lightweight** - The quantity of energy contained in a battery is proportional to its weight, which is measured by its energy density. Solid-state batteries are estimated to have a 2.5-fold higher energy density than contemporary lithium-ion batteries. Because of the massive improvement in the energy density of solid-state batteries, they will be more smaller and lighter. Batteries with a higher energy density could be substantially lighter while storing the same amount of energy. For mobile power applications, weight is a major consideration, thus this could be a game-changer. Vehicles

¹⁴ [What is a Solid-State Battery And Why You Will Want One | Dragonfly Energy](#)

¹⁵ [What is a Solid-State Battery And Why You Will Want One | Dragonfly Energy](#)

such as cars, trucks, RVs, boats, and planes could all benefit from being lighter. Batteries with a higher energy density could be substantially lighter while storing the same amount of energy. For mobile power applications, weight is a major consideration, thus this could be a game-changer. Vehicles such as cars, trucks, RVs, boats, and planes could all benefit from being lighter. Electric vehicles have a lot to gain from this technology because they can achieve a lot more range with a lot less weight and battery storage space.

2. **Faster Charging Times** - Solid-state batteries have the ability to perform at extremely high rates of power. According to research, they may be able to recharge 4-6 times faster than current technology while remaining safe.
3. **Safer Battery Experience** - Liquid electrolytes used in lithium-ion batteries are highly flammable and volatile. These electrolytes must also be kept away from air. Solid-state batteries don't have any liquid parts, hence this volatile component isn't present.

While lithium-ion batteries are susceptible to phenomena such as thermal runaway, which can result in an explosion and fire, the solid electrolytes used in solid-state batteries are nonflammable, posing a lesser danger of fire. While the battery will heat up, it contains nothing flammable that will catch fire. The cherry on top is that these batteries will have fewer safety features than lithium-ion batteries. Eliminating extra circuitry can result in smaller, lighter battery packages, increasing the energy density of solid-state batteries even more.

In some cases, these batteries may not even require an external BMS, considerably simplifying their construction.¹⁶

1.3 How a Solid-State Battery Works

The anode, cathode, and electrolyte are the three main components of any battery. A battery functions because charged ions desire to move through the electrolyte from the cathode to the anode. This occurs because the battery's carefully selected components generate a chemical process that releases free electrons. As a result, a positive charge accumulates on the cathode of the battery. This draws the anode's negatively charged free electrons. Those liberated electrons seek to go from the anode to the cathode. They provide electricity to your gadget as they do so.

¹⁶ [What is a Solid-State Battery And Why You Will Want One | Dragonfly Energy](#)

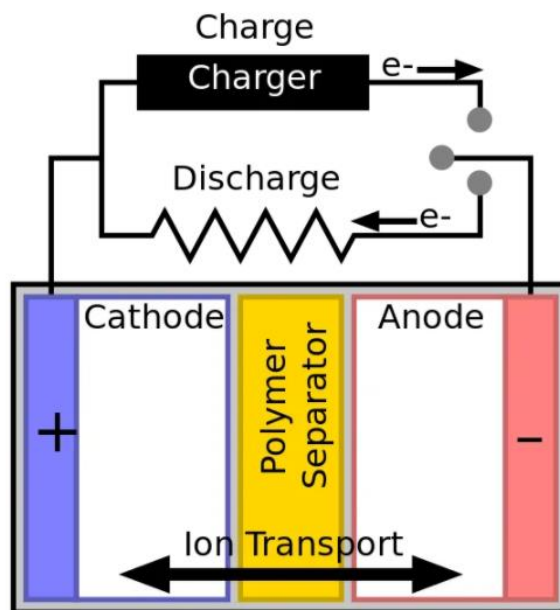


Figure SC-BAT-SRB- 4: Solid State Battery Diagram

Keep in mind that electrical and chemical forces are continually trying to balance each other out. Consider a fully charged battery like a seesaw with one axis slanted in one way. The charges want to descend this plane until the reaction is balanced. Recharging returns all of the ions to their original places. Consider charging as a way to increase the speed of our unbalanced seesaw from the previous metaphor.

1.4 Challenges

- **Cost** - Solid-state batteries are typically expensive to manufacture and use difficult-to-scale production procedures that necessitate the use of expensive vacuum deposition equipment. As a result, consumer-based applications become prohibitively expensive. In 2012, it was projected that a 20 Ah solid-state battery cell would cost US\$100,000, and that a high-range electric car would require between 800 and 1,000 of these cells, based on then-current technology. Similarly, the high cost of solid-state batteries has hampered their acceptance in other areas, such as cellphones.
- **Temperature and pressure sensitivity** - Low-temperature procedures can be difficult. Solid-state batteries have had a bad track record in the past. To keep contact with the electrodes, solid-state batteries with ceramic electrolytes require a lot of pressure. Mechanical stress can cause solid-state batteries with ceramic separators to fail.
- **Dendrites** - The anode's lithium metal dendrite pierces the separator and grows towards the cathode. For increased energy densities, safety, and faster recharging periods, solid lithium (Li) metal anodes in solid-state batteries are replacement options in lithium-ion batteries. The creation and proliferation of Li dendrites is a problem for such anodes. Short circuits are caused by dendrites penetrating the separator between the anode and the cathode. Overheating occurs, which can lead to fires or explosions due to thermal runaway. Coulombic efficiency is reduced by Li dendrites. During charge and discharge, dendrites are typically

formed during electrodeposition. As the battery charges, lithium ions mix with electrons at the anode surface, generating a layer of lithium metal. Lithium should be deposited evenly on the anode in the ideal situation. Dendrites, on the other hand, occur when growth is uneven. The most effective technique for reducing dendrite formation and improving cycle performance was determined to be stable solid electrolyte interphase (SEI). Although this remains hypothetical, solid-state electrolytes (SSEs) may limit dendritic development. Nanoporous ceramic separators that restrict Li dendrite formation up to critical current densities were discovered in a 2018 study.

SAMPLE TOPIC ACTIVITIES

Activity 1: Group Work/Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research different solid-state batteries of their interest and provide an explanation of their working mechanism. A part of a Groups presentation should show the difference between solid state and liquid-electrolyte battery.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Topic 2: Solid Electrolyte

2.1 Electrolyte

Organic and inorganic electrolytes are the two types of solid electrolytes. Because inorganic electrolytes are non-flammable and do not include harmful compounds, they have a higher level of safety. As a result, the focus of this pamphlet is on inorganic electrolytes. The diagram below provides an overview of the material categories as well as a list of individual structural kinds.¹⁷

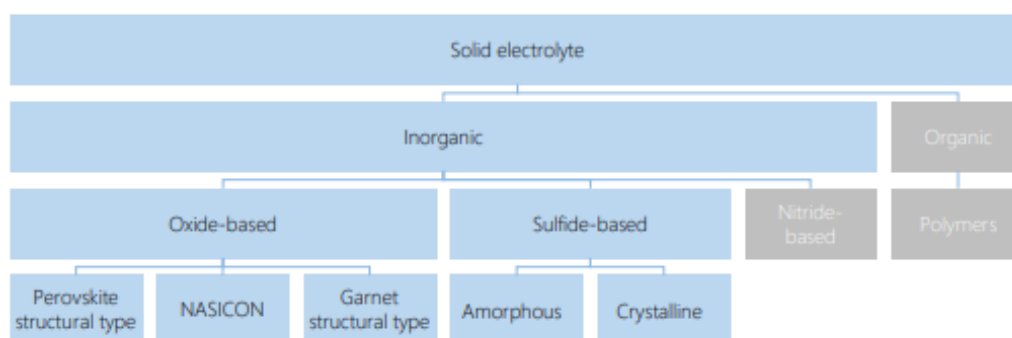


Figure SC-SRB-3: Categories of types of Electrolytes. (Source: https://www.pem.rwth-aachen.de/global/show_document.asp?id=aaaaaaaaabdqubuc)

- **Oxide-based electrolytes** are usually chemically stable and work well with high-energy cathode materials. The ion conductivity is, however, lower than that of sulfide-based electrolytes.
 - Materials with a perovskite or garnet structure, as well as NASICON, are promising among oxide-based electrolytes. The properties are classified qualitatively in the diagram below. LiPON, LISICON, lithium halides, and lithium hydrides are more examples.

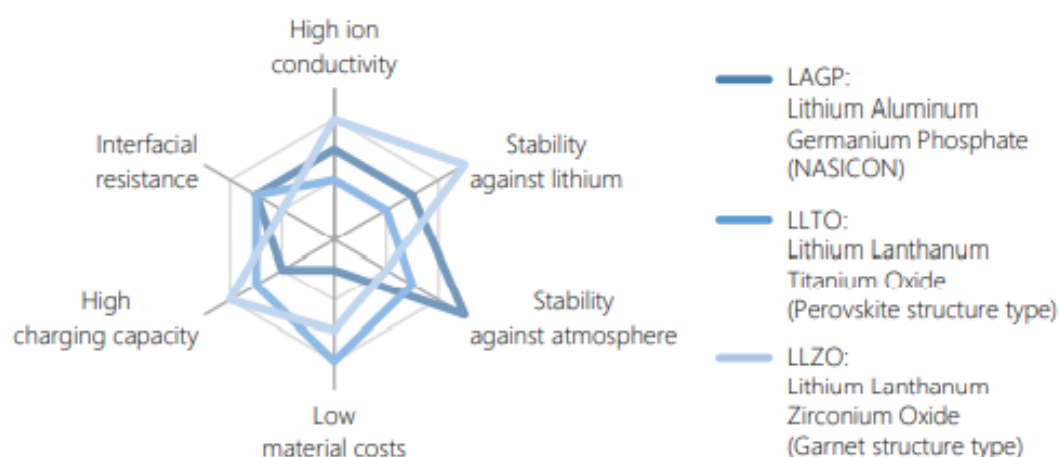


Figure SC-BAT-SRB- 5: qualitative classification of the properties.

¹⁷ [Masterdokument_Zellherstellung_v63_Vordruckfreigabe.pptx \(rwth-aachen.de\)](#)

- **Sulfide-based electrolytes** generally have a higher ionic conductivity, but are more chemically unstable.
 - At ambient temperature, amorphous lithium tin phosphorus sulfide (LSPS) is cheap and has a very high ion conductivity. Incompatibilities with lithium metal, on the other hand, constitute an issue.
 - Lithium-germanium-phosphorus sulfide (LGPS) crystalline ion conductivity is comparable to that of liquid electrolytes. LGPS, on the other hand, is extremely susceptible to moisture and lithium metal, and germanium is extremely expensive.

2.2 Materials of an all-solid-state battery

Anode

- Lithium metal anodes are suitable because they have the largest theoretical capacity for achieving the highest energy density. Metallic lithium, on the other hand, has a tendency to produce dendrites, which the solid electrolyte must prevent. Furthermore, lithium produces a passivating layer with ambient oxygen, necessitating handling in an inert atmosphere.
- Silicon as an anode material promises excellent energy densities, but it endures drastic volume changes during lithium alloy production..

Cathode

- A metal oxide is used as the cathode. Because there are essentially no materials that have been specifically created for the all-solid-state Battery, known cathode materials are employed in most cases.
- In theory, a wide range of proven cathode materials can be employed, depending on the electrolyte, ranging from affordable and safe materials like lithium iron phosphate (LFP) through lithium nickel manganese cobalt oxide (NMC). Only lithium cobalt oxide (LCO) as a cathode material and LLZO as an electrolyte exhibit sufficient stability and performance in practice.

2.3 Chances and Challenges

- **High energy densities** of up to 500 Wh/kg and 1000Wh/L at cell level are possible with a metallic anode and a thin electrolyte layer.
- A **high level of safety** can be guaranteed during operation. The risk of explosion and fire is minimized as no flammable liquids or gases can escape.
- **Short charging times** can be achieved because the solid electrolyte greatly reduces the risk of dendrite formation.
- The **operational capability** is supplied throughout a wide temperature range of approximately -30 to +100°C.
- The operational capability is supplied throughout a wide temperature range of approximately -30 to +100°C.
- The **material compatibility** between the solid electrolyte and the lithium metal anode must be ensured by appropriate changes, as well as the resistance at the electrolyte-electrode contact.

SAMPLE TOPIC ACTIVITIES

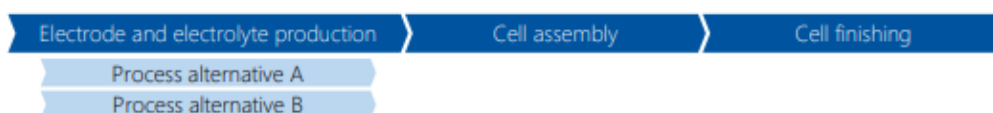
Activity 1: Group Work/Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research innovative solid-state materials that they would use to design a solid-state battery. They should provide a conceptual design of what they think the battery would look and work like. Students should also study how batteries are packed and why they are packed that way, propose a different packing arrangement and justify that packing arrangement.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Topic 3: Manufacturing of Solid Batteries

The creation of an all-solid-state Battery can be separated into three overall steps: Electrode and electrolyte production, cell assembly, and cell finishing. A generally valid process chain does not exist; instead, a vast variety of alternate process chains may be implemented. These differ from the lithium-ion battery production method in some ways. This item compares and contrasts two different process options that differ primarily in electrode and electrolyte production:



The manufacturing of pouch cells with inorganic solid electrolytes is the subject of both process options. For all-solid-state batteries, the pouch cell format looks to be the best suitable:

1. **Round or prismatic cell** - Due to the solid components of an all-solid-state battery, windings pose significant complications. Ceramic layers that are fragile can develop cracks. Furthermore, the issue of sufficient layer adhesion has yet to be resolved.
2. **Pouch Cell** - Because the flat layers are not distorted, stacking is useful for all-solid-state batteries. Furthermore, the layer compound is generated throughout the electrode and electrolyte manufacturing processes, leaving only the basic cells to be stacked later.



Figure SC-BAT-SRB- 6: Prismatic Cell and Pouched Cell

A dry room is essential for production due to the reactivity of the materials with the atmosphere. An inert environment, such as argon, is preferable when handling metallic lithium. A qualitative assessment of the transferability of expertise developed in the production of lithium-ion battery cells is done for each process stage.

3.1 Manufacturing Process: Electrode and Electrolytic Production – Alternative A

The composite of cathode, electrolyte, and anode is created in electrode and electrolyte manufacture. An elementary cell exists after electrode and electrolyte synthesis. The major feature of the first process chain, Process Chain A, is a continuous extrusion process in which the layers are then laminated. This manufacturing chain is particularly well suited to all-solid-state sulfide batteries.

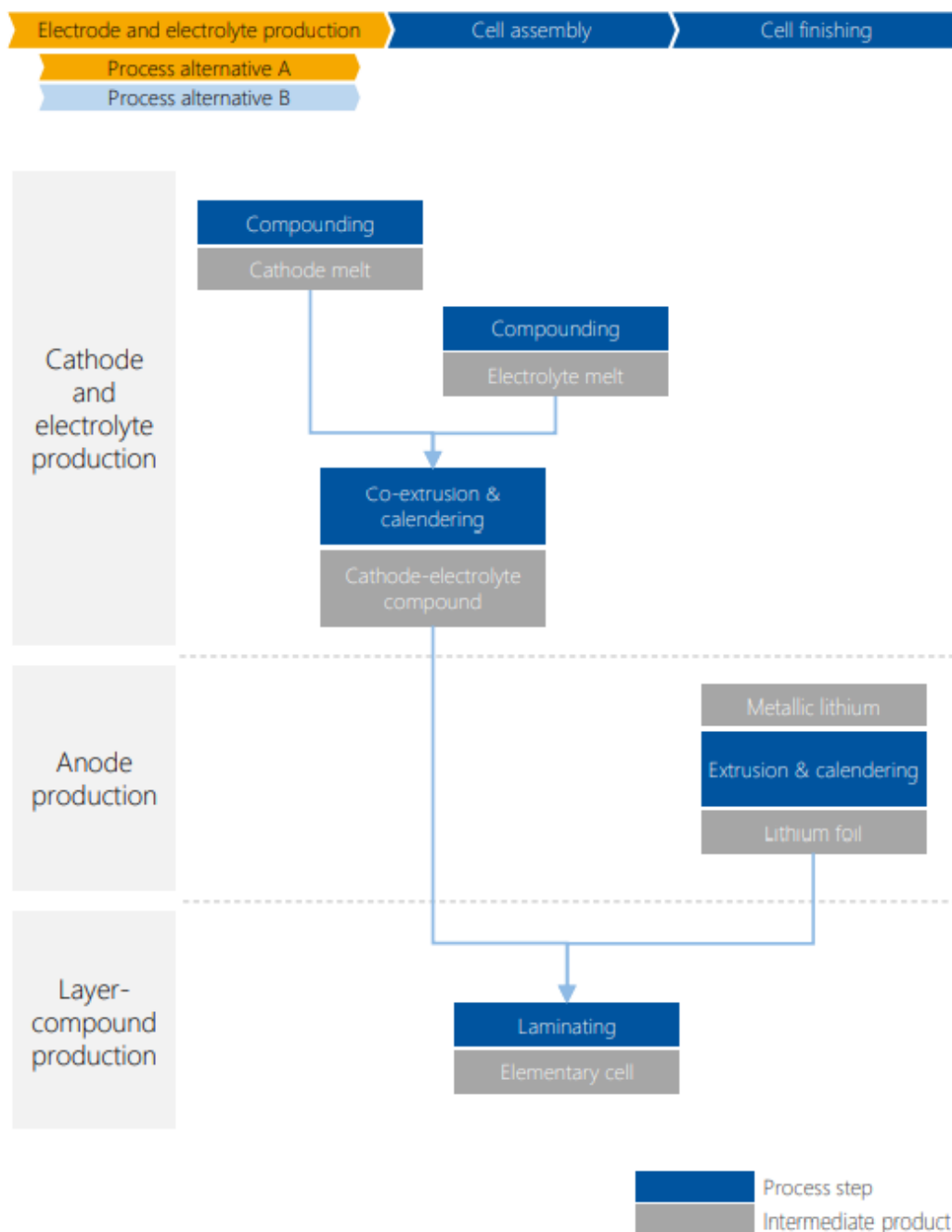


Figure SC-BAT-SRB- 7: Schematic Diagram of Manufacturing Process Alternative A (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

3.1.1 Compounding: Cathode and Electrolyte Production

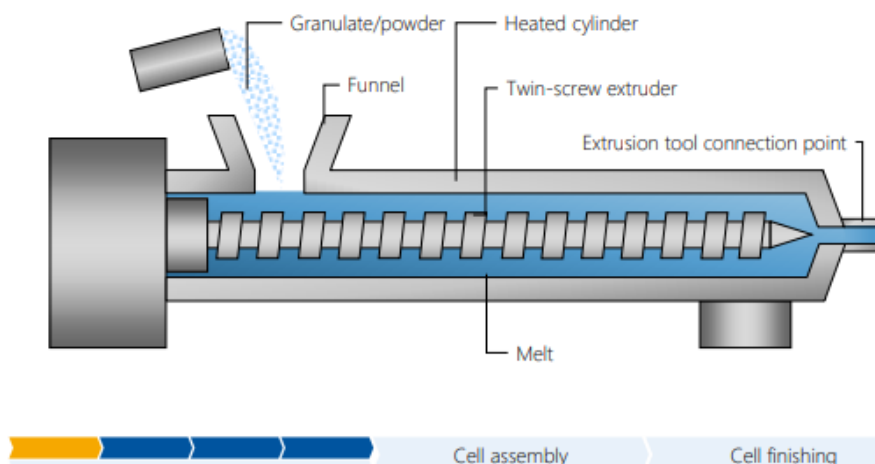


Figure SC-BAT-SRB- 8: Cathode and Electrolyte Production (Source: [Masterdokument Zellherstellung_v63 Vordruckfreigabe.pptx \(rwth-aachen.de\)](#))

3.1.1.1 Production process

Two distinct compounding procedures generate the cathode and electrolyte melts. The material components are put into a twin-screw extruder's heated barrel and can be delivered as granulate or powder. The extruder's rotational movements introduce energy into the material components. As a result, the melt is homogeneous. Electrolyte particles, which minimize the resistance between the cathode and the electrolyte, as well as binders and additives, are mixed with the cathode active material.

3.1.1.2 Additional information

The material components of the electrolyte are electrolyte particles and polymer binders.

Process Requirements - Feeding quantity of the individual materials - Cylinder temperatures and pressures - Speed and torque of the extruder - Shear Energy	Parameters and	Quality Features (Excerpt) - Homogeneity of the melt - Viscosity of the melt - Number and size of agglomerates
Challenges (Excerpt) Homogenous mixing of the individual materials		Technology Alternatives (Excerpt) High-performance mixing plant

3.1.1.3 Co-Extrusion: Cathode and Electrolyte Production

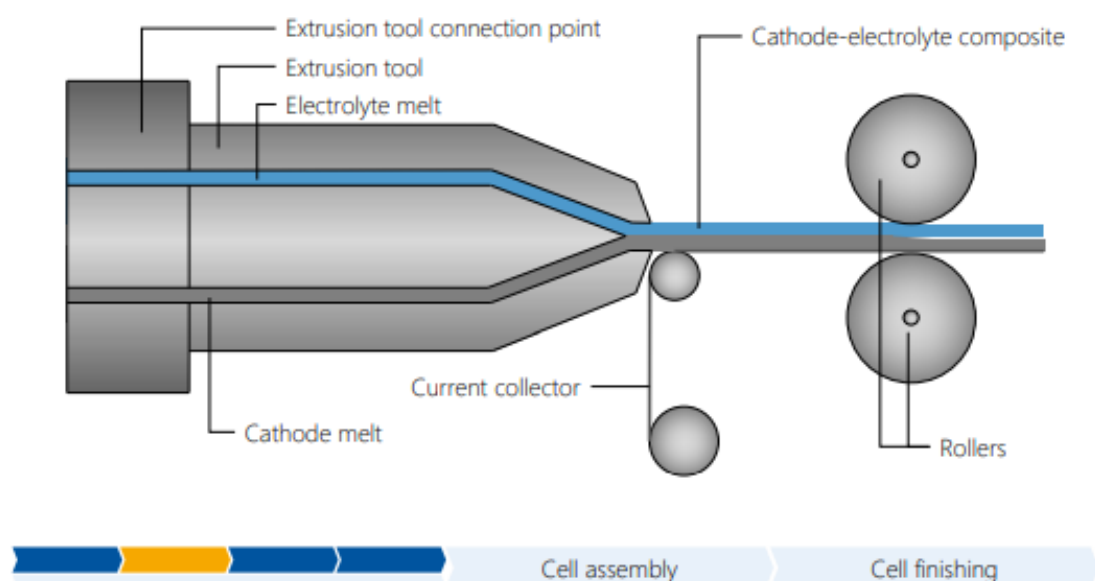


Figure SC-BAT-SRB- 9: Co-Extrusion ((Source: [Masterdokument_Zellherstellung_v63_Vordruckfreigabe.pptx \(rwth-aachen.de\)\)](#))

Production process

In a suitable die, the cathode and electrolyte melts are co-extruded. This results in a cathode and electrolyte layer composite. Separate tubes feed cathode and electrolyte melts via the extrusion die. The melts travel via the channels to the extrusion die's exit. The melts are extruded onto a current conductor using a slot die. The bond is calendared after extrusion to increase adhesion between individual layers and achieve specified layer thicknesses.

Process parameters & requirements • Adjustment of the layer thickness • Melt feed rate • Temperature • Pressure • Roll speed • Pressing pressure of the calender rolls	Quality features [excerpt] • Coating thickness • Layer width • Adhesion between layers
Challenges [excerpt] • Different process temperatures depending on the materials	Technology alternatives [excerpt] • Foil casting (tape casting) • Screen printing
Transferability of lithium-ion battery cell manufacturing expertise	

3.1.1.4 Extrusion and Calendaring: Anode Production

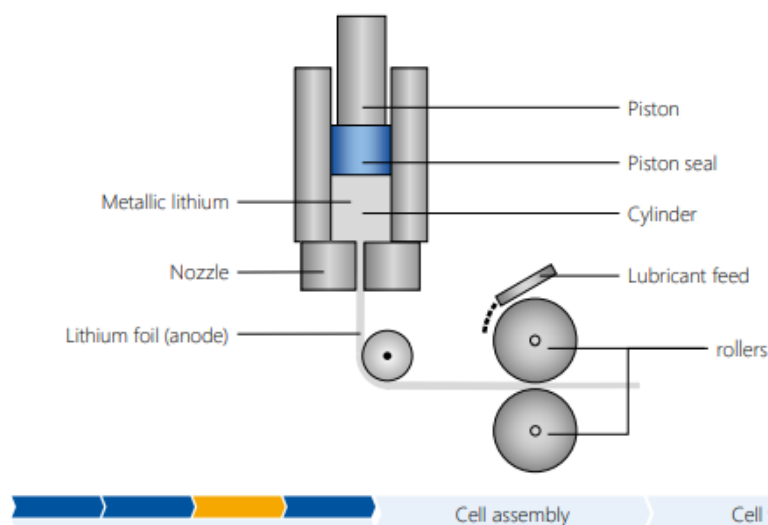


Figure 4: Extrusion and Calendaring for Anode Production (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

An all-solid-state battery's anode can be made of metallic lithium foil. Extrusion with subsequent calendaring can be used to make this lithium film. Liquid lithium is poured into the cylinder of a piston extruder for this purpose. The lithium is then forced into a nozzle by a piston. Calendaring after extrusion ensures homogeneity and the optimum film thickness. The film is rolled under pressure by two rollers with the addition of a lubricant for this purpose.

Additional information

The rollers must be able to work with lithium's adhesive properties. Polymer-coated rollers, such as those composed of polyacetal, can do this.

Process parameters & requirements - Extrusion speed - Temperature - Nozzle geometry - Pressing pressure of the calender rolls - Supply speed of the lubricant - Roll speed	Quality features [excerpt] - Film thickness - Foil width - Homogeneity of the lithium foil
Challenges [excerpt] Adhesion tendency of metallic lithium during calendaring	Technology alternatives [excerpt] - PVD process - Atomic layer deposition (CVD process)
Transferability of lithium-ion battery cell manufacturing expertise	

3.1.1.5 Laminating: Layer Compound Production

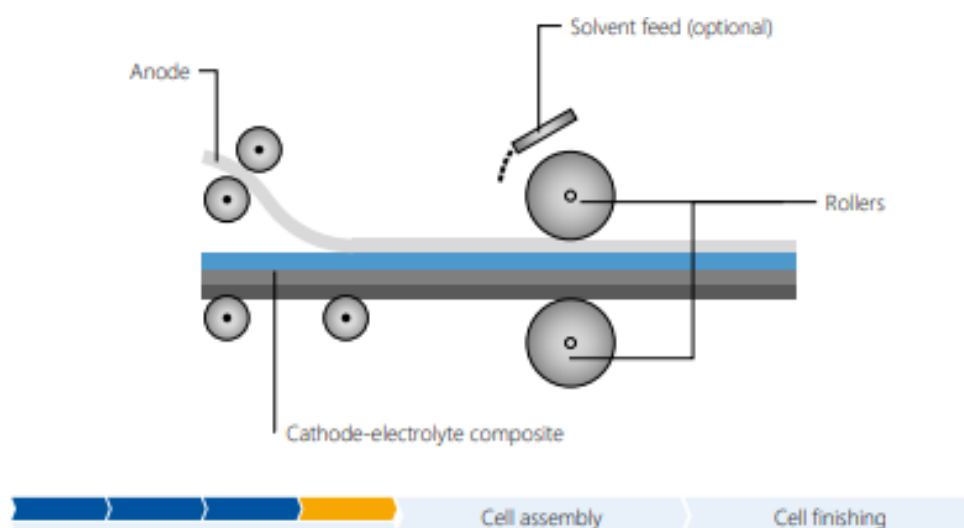


Figure SC-BAT-SRB- 10: Laminating for Layer Compound Production (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

The lithium foil is laminated onto the cathode-electrolyte composite once it has been made. The two layers are brought together using rollers for this reason. Two rollers are used to push the two layers together in the next stage. To achieve larger adhesion forces, they are heated. Polymers penetrate from one layer to the next during heating and pressing, producing the link between the anode and the electrolyte. supplementary information There is a difference between dry and wet lamination. Wet lamination involves moistening the contact surfaces with a solvent prior to lamination. This enables for reduced temperature and pressure lamination.

Process parameters & requirements - Feeding speed of the layers - Roll speed - Pressure (order of magnitude 12-40 N/mm², adjustable via the roll pressure or the calender gap) - Optional heating of the layers	Quality features [excerpt] - Adhesion between layers - Desired composite thickness - Geometry of the composite
Challenges [excerpt] Adhesion tendency of metallic lithium during calendaring	Technology alternatives [excerpt] Pressing and subsequent sintering

Transferability of lithium-ion battery cell manufacturing expertise

3.2 Manufacturing Process: Electrode and Electrolytic Production – Alternative B

The physical vapor deposition (PVD) process, in which the different layers are applied one after the other, is the main element of the process chain B shown below. This manufacturing chain, which depicts the production phases of a thin-film cell, is especially useful for oxide-based all-solid-state batteries.

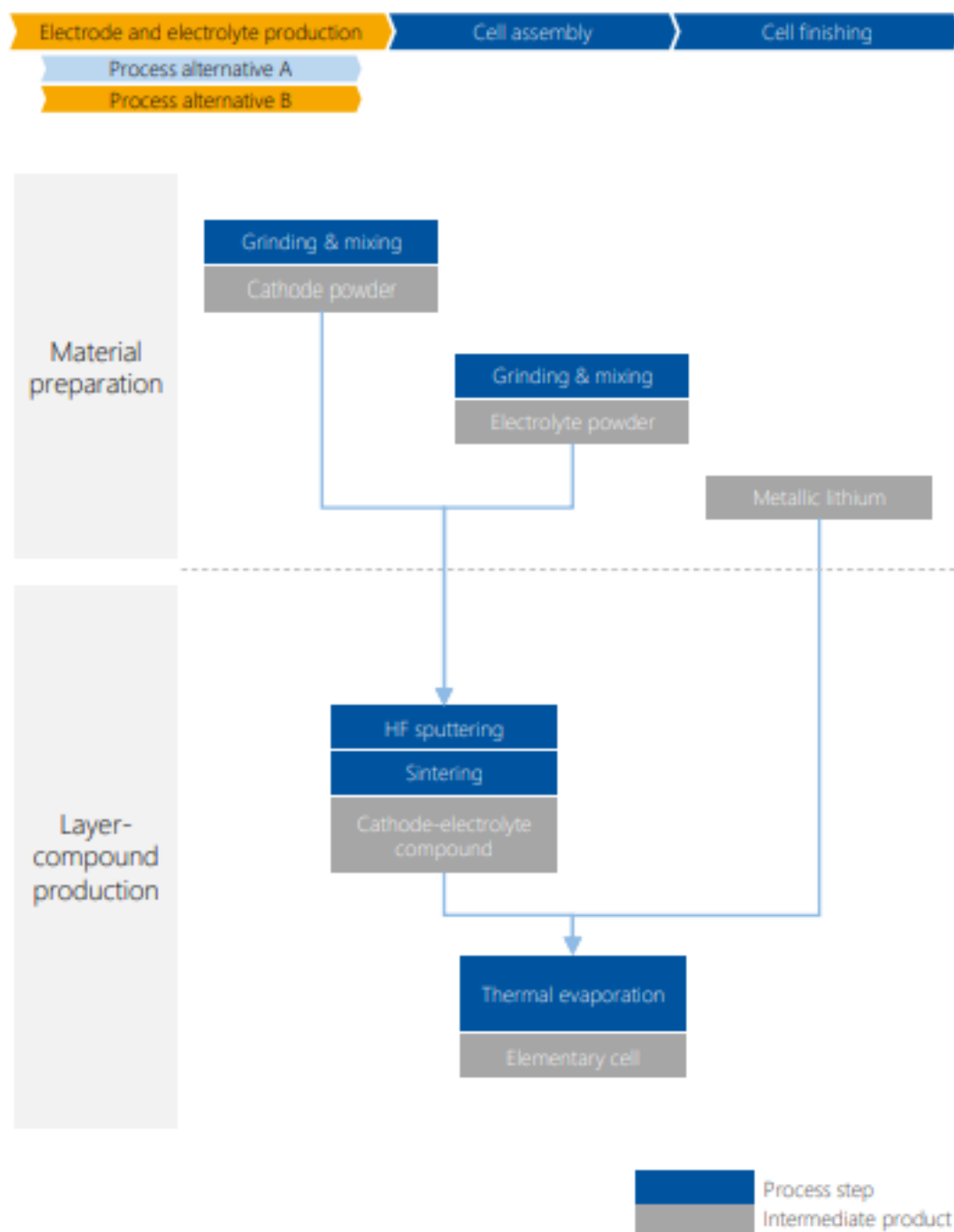


Figure SC-BAT-SRB- 11: Schematic of Manufacturing Process Alternative B (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

3.2.1 Grinding and Mixing

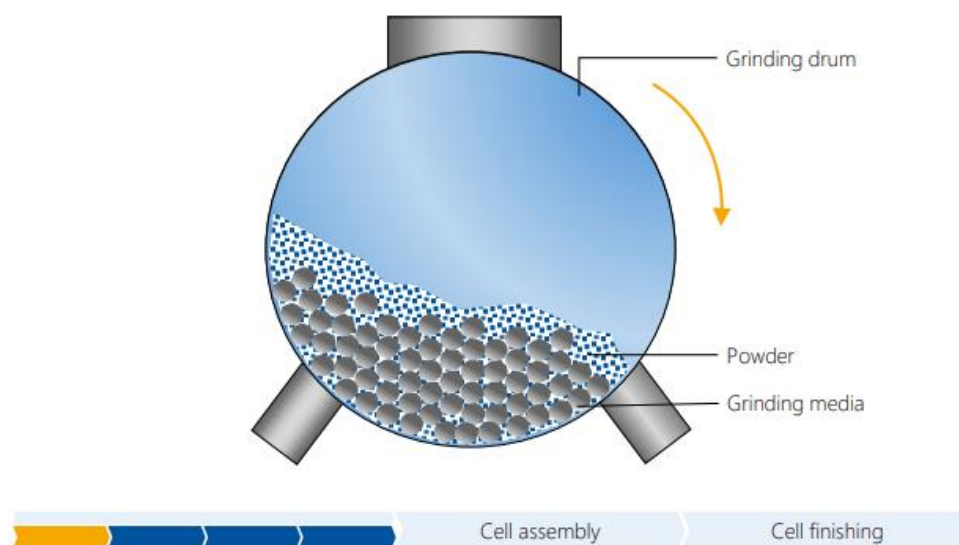


Figure SC-BAT-SRB- 12: Grinding and Mixing (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

3.2.1.1 Production process

In a ball mill, the cathode powder and the electrolyte powder are made separately. The corresponding starting materials are fed onto a cylindrical grinding drum for this purpose. There are balls employed as grinding media in this grinding drum. The rotational movements of the cylinder mix the beginning components. Furthermore, the circular movement ensures that the grinding media and the starting material move relative to each other while the latter is ground. After that, the powder is calcined to achieve the necessary powder characteristics.

Process parameters & requirements • Ball material • Speed • Grinding time • Cylinder material • Quantity of starting materials	Quality features [excerpt] • Average powder particle size • Homogeneity of the powder (degree of mixing)
Challenges [excerpt]	Technology alternatives [excerpt] • Sol-gel process
Transferability of lithium-ion battery cell manufacturing expertise	

3.2.1.2 Layer Compound Production: High Frequency Sputtering

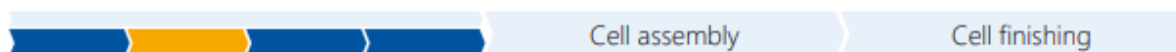
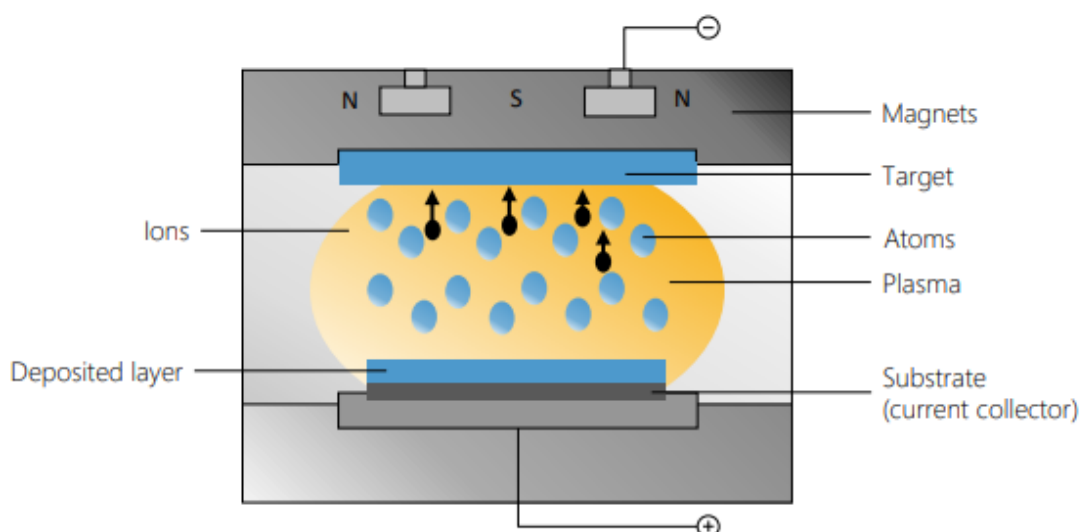


Figure SC-BAT-SRB- 13: High Frequency Sputtering (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

High-frequency sputtering is used to create the cathode and electrolyte layers from cathode and electrolyte powders. The target for the sputtering technique is first made from powders using a die or hot pressing method. The current collector also serves as the process's substrate. The cathode layer is deposited in the first step. An electrolyte layer is next put on top of the cathode layer. Ions are fired at the target of the sputtering process. Atoms are knocked out of the target in this process, which then enter the gas phase and accelerate to the substrate. The layer is thus grown up atom by atom on the substrate's surface.

Additional information: High-frequency sputtering takes place in a vacuum chamber

Process parameters & requirements • Temperature • Deposition time • Process pressure • Ambient atmosphere • Process power/power density • Target diameter & target distance	Quality features [excerpt] • Layer thickness of the current collector • Layer thickness of the cathode and the electrolyte
Challenges [excerpt] • Slow process with low throughput • Low layer thicknesses lead to low energy densities	Technology alternatives [excerpt] • Chemical Vapor Deposition
Transferability of lithium-ion battery cell manufacturing expertise	

3.2.1.3 Layer Compound Production: Sintering

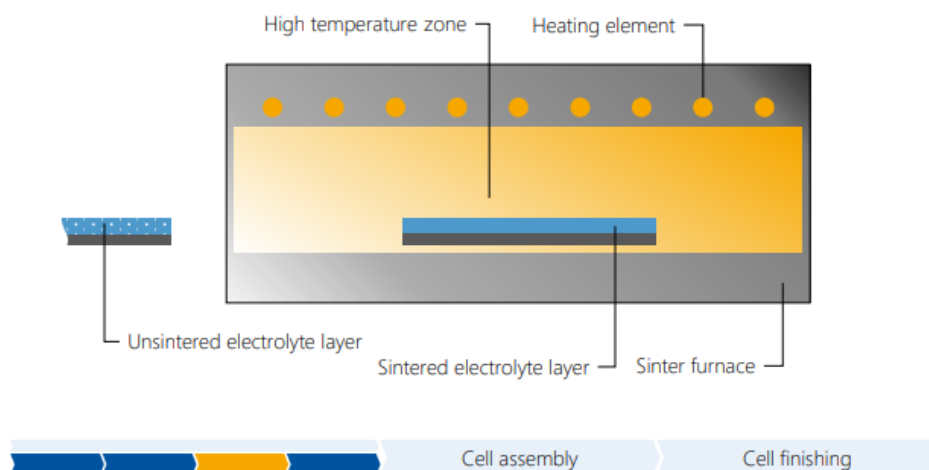


Figure SC-BAT-SRB- 14: Sintering (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

The cathode and electrolyte layers are compressed during sintering. As a result, by enhancing the link between the two layers, the resistance at the interface electrolyte – electrode can be lowered. A sintering furnace is used to sinter the cathode-electrolyte complex. The material is heated to just below its melting point. The final porosity of the materials can be modified based on the process parameters chosen.

Additional information

To avoid reactions with the environment, the sintering process takes conducted in an inert atmosphere or in a vacuum. To produce a suitably low interfacial resistance, sintering is especially important for oxide-based solid electrolytes.

Process parameters & requirements • Sintering temperature (800°C - 1000°C for high temperature sintering, up to 1300°C depending on material) • Sintering pressure (sintering at atmospheric pressure preferred) • Sintering time (usually several hours)	Quality features [excerpt] • Compound adhesion • Porosity • Interfacial resistance
Challenges [excerpt] • Compatibility of the sintering temperatures of different materials • High energy input for high sintering temperatures	Technology alternatives [excerpt] • Laser-based processes
Transferability of lithium-ion battery cell manufacturing expertise	

3.2.1.4 Layer Compound Production: Thermal Evaporation

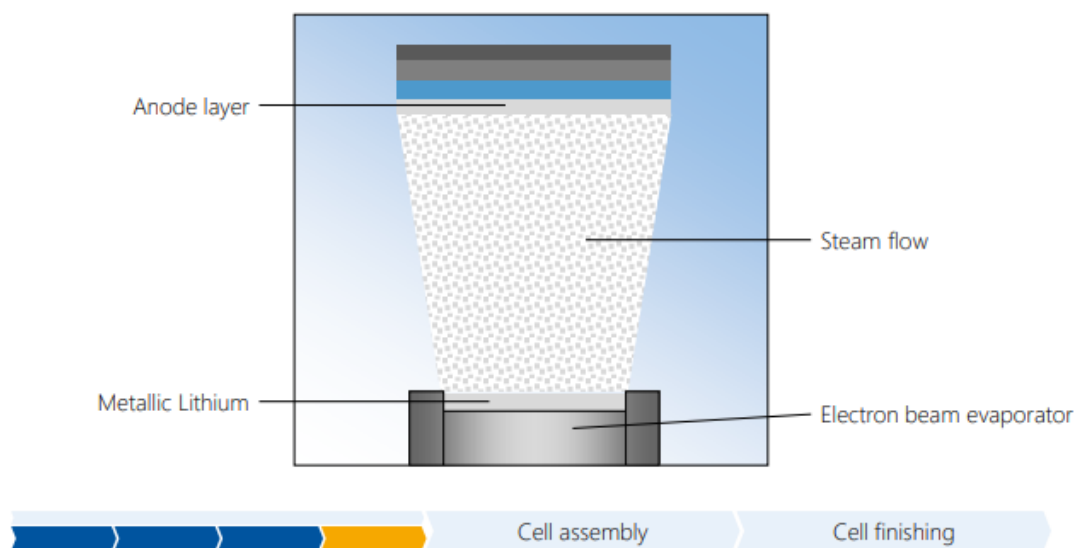


Figure SC-BAT-SRB- 15: Thermal Evaporation (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

The anode can be applied to the cathode – electrolyte complex using thermal evaporation. The anode material is made of metallic lithium. Thermal evaporation involves heating metallic lithium to temperatures around boiling point, such as with an electron beam evaporator, so that it can enter the vapor phase. In the vacuum chamber, the steam distributes evenly. Condensation on the surface of the electrolyte, which has a lower temperature, forms the layer.

Additional information

Like sputtering, thermal evaporation takes place in a vacuum chamber

Process parameters & requirements • Current • Resistance • Deposition time • Process pressure and temperature • Condensation rate • Distance to evaporation source	Quality features [excerpt] • Layer thickness of the anode • Layer purity
Challenges [excerpt] • Long cycle times • Low layer adhesion	Technology alternatives [excerpt] • CVD process

Transferability of lithium-ion battery cell manufacturing expertise

3.3 Manufacturing Process: Cell Assembly and Cell Finishing

The battery cell is constructed from existing elementary cells during cell assembly. In comparison to traditional lithium-ion battery manufacture, the all-solid-state Battery's cost- and time-intensive formation and aging can be simplified. Furthermore, electrolyte filling and degassing are no longer required. For grading and quality assurance, the battery cell is put through a series of tests.

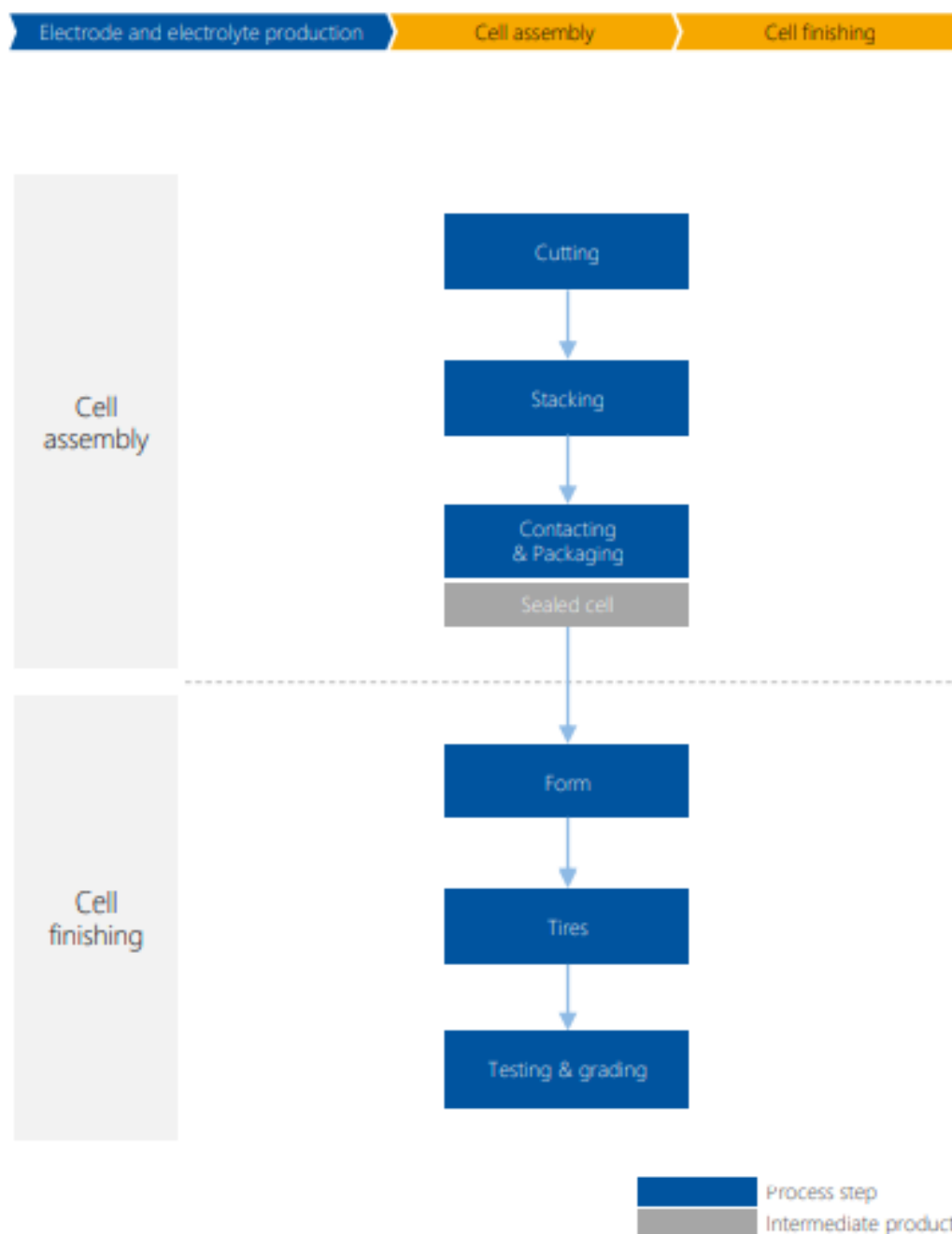


Figure SC-BAT-SRB- 16: Schematic of the Manufacturing process of Cell Assembly and Finishing (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

3.3.1 Cell Assembly: Cutting

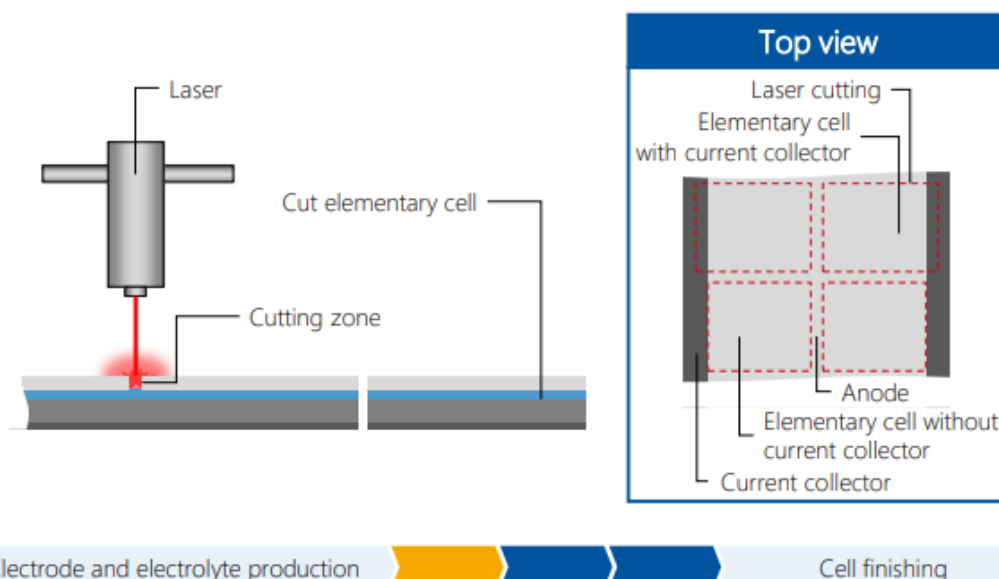


Figure SC-BAT-SRB- 17: Cutting (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

The elementary cells must be cut to size once the layer lamination is completed so that they can be stacked in the next process cut. Laser cutting can be used to accomplish this. The laser-cutting procedure, for example, can be done with a fiber laser. With the correct energy input into the layer compound, a burr-free cut can be obtained. It's also vital to avoid melting the corresponding layer materials, which could result in an electrical connection between the layers and hence a short circuit.

Process parameters & requirements • Cutting speed (e.g. 100 mm/s) • Spot size (e.g. 25 μm) • Pulse energy and duration (e.g. 30 ns) • Pulse repetition rate (e.g. 100 kHz) • Laser power or energy per unit length • Wavelength (e.g. 1064 nm)	Quality features [excerpt] • Heat input • Dimensions • Burr-free edges
Challenges [excerpt] • Mutual contamination of the layers • Different laser settings due to different materials	Technology alternatives [excerpt] • Punching
Transferability of lithium-ion battery cell manufacturing expertise	

3.3.2 Cell Assembly: Stacking

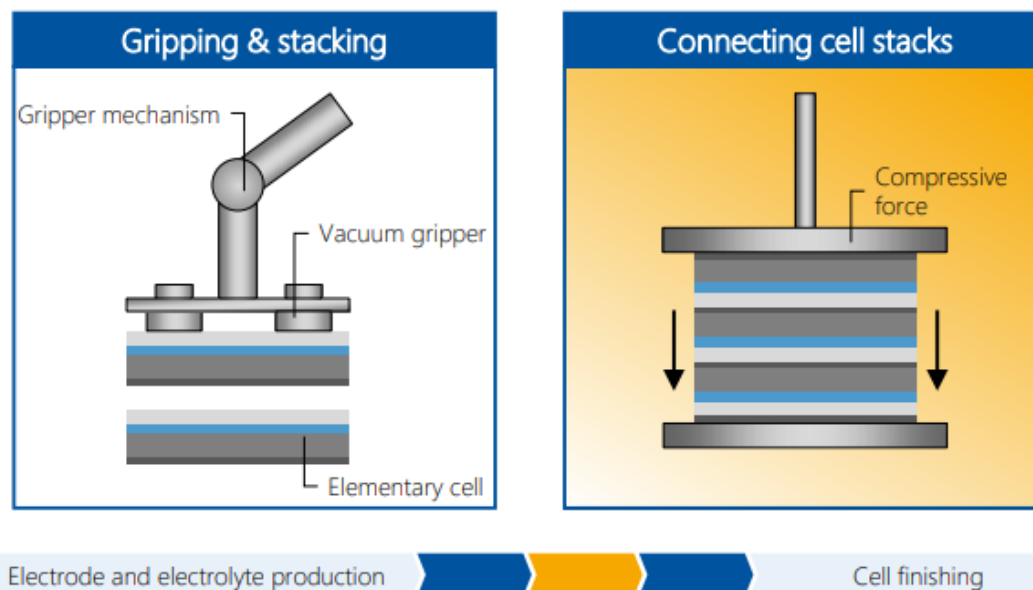


Figure SC-BAT-SRB- 18: (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de)).

Production process

In a bipolar design, the elementary cells can be stacked directly. The cells are connected in series, resulting in a doubled cell voltage. With grippers, the elementary cells are stacked on top of one another. Make sure the surface is clear of damage before choosing a grasping technique. A cell's completion necessitates the inclusion of a current collector. Pressure and heat are subsequently used to laminate the elementary cells together. A bipolar cell stack is the end outcome.

Additional information

Further advantages of direct stacking lie in its simplicity and effectiveness. Additional packaging costs are minimized.

Process parameters & requirements - Stacking speed - Positioning accuracy - Pressure - Temperature	Quality features [excerpt] - Positioning accuracy - Damage-free stacking
Challenges [excerpt] Long process times	Technology alternatives [excerpt] Winding the elementary cells. Due to the brittle solid electrolyte, however, winding all-solid-state batteries requires a more sophisticated process.
Transferability of lithium-ion battery cell manufacturing expertise	

3.3.3 Cell Assembly: Contacting and Packaging

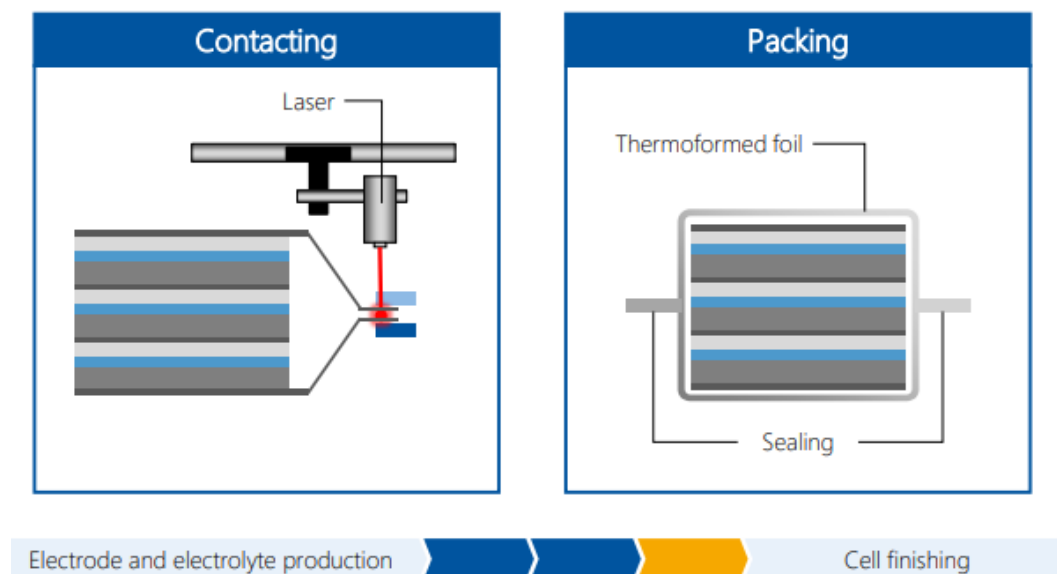
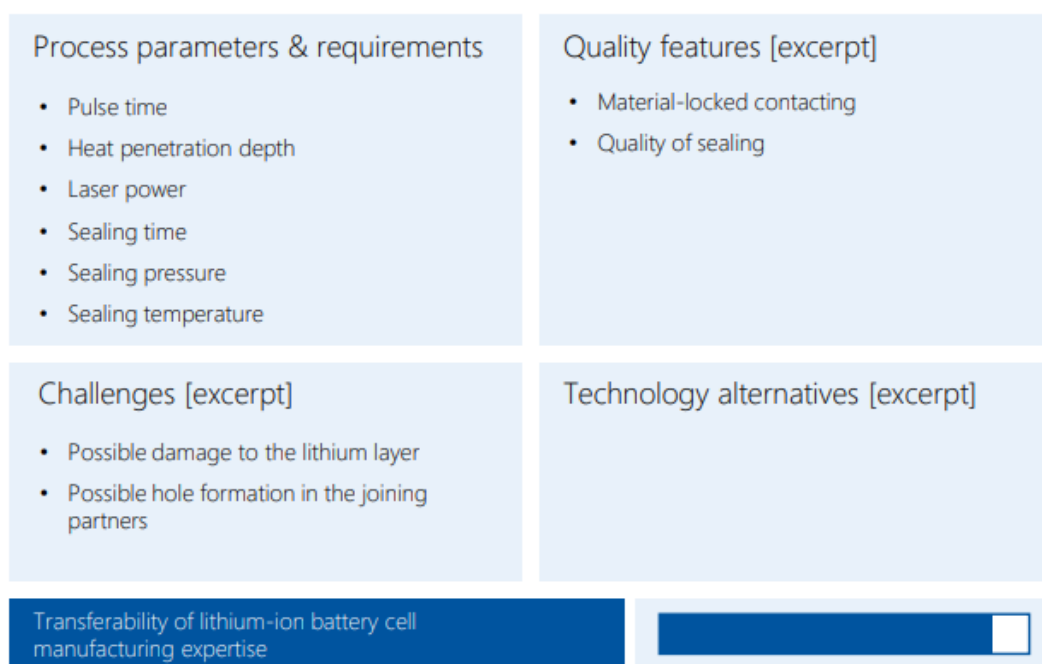


Figure SC-BAT-SRB- 19: Contacting and Packaging (Source: [Masterdokument Zellherstellung v63 Vordruckfreigabe.pptx \(rwth-aachen.de\)](#))

Production process

Only the two outer current collectors in the bipolar cell stack must be touched with the contact tabs. For current collectors, it may be able to employ less expensive materials. Laser beam or ultrasonic welding can be used to make contact. The all-solid-state Battery cell is sealed after contacting and placed in an electrically insulated package to protect it from environmental impacts. Packaging materials comprised of metal-plastic mixture foils are acceptable. External current collectors must be electrically shielded before being put into the foil. Using impulse or touch sealing, the packaging is entirely sealed.



3.3.4 Cell Finishing: Formation

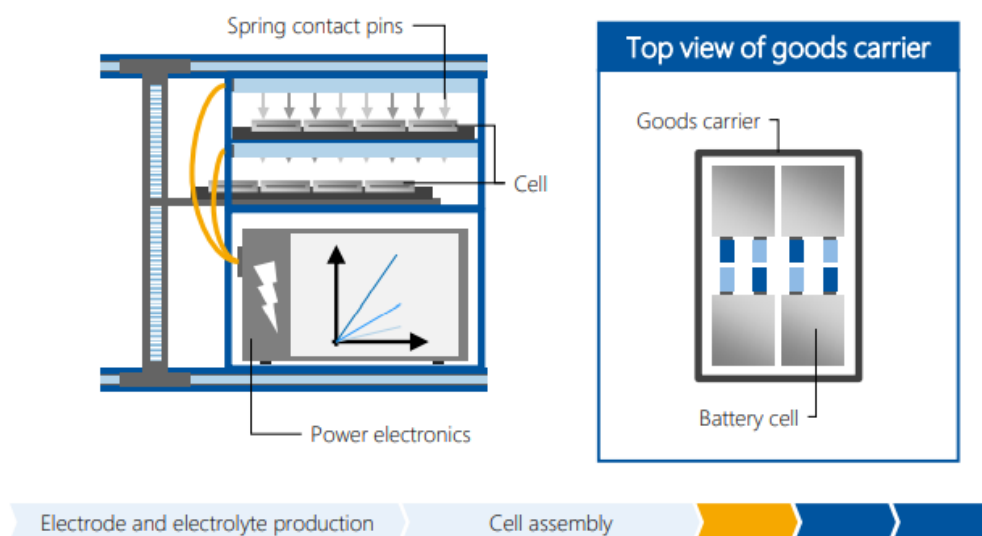


Figure SC-BAT-SRB- 20: Formation (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

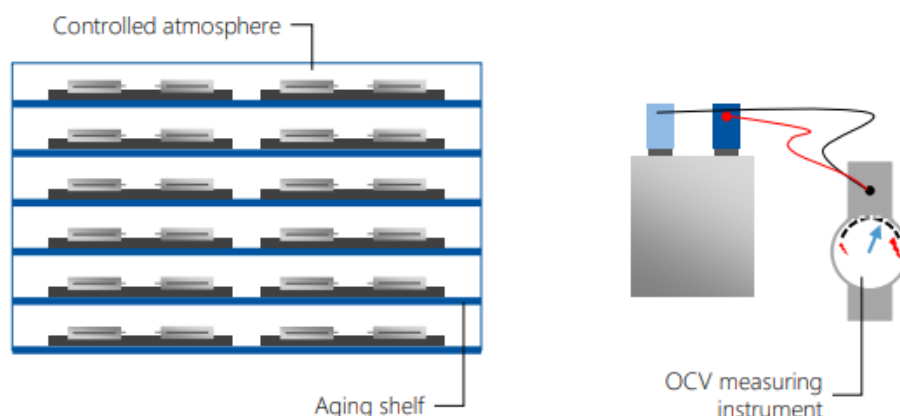
The battery cell is exposed to the first charging and discharging cycles during creation. An all-solid-state battery with a lithium metal anode is already charged when it is constructed. Spring contact pins contact the cell in forming racks.

Additional information

Between the electrolyte and the electrodes, an interface layer layer. This layer has a substantial impact on ion conductivity and, as a result, cell performance. However, in comparison to lithium-ion batteries, this layer is thinner and entirely produced after only a few cycles. The capacity and internal resistance of the battery cell stay relatively consistent even after one discharge/charge cycle. When compared to lithium-ion cells, the formation time and cost are significantly reduced.

Process parameters & requirements • Defined C-rate for first discharging and charging cycle • Gradual increase of the C-rate • Current and voltage profile • Low contact resistances at the spring contact pins	Quality features [excerpt] • Formation of the interface layer • Internal resistance of the cell
Challenges [excerpt] • Location of cells • Contacting method • Process temperature	Technology alternatives [excerpt] • Formation is largely based on extensive experiential knowledge, which does not yet exist for all-solid-state batteries due to lack of series production maturity.
Transferability of lithium-ion battery cell manufacturing expertise	☐

3.3.5 Cell Finishing: Aging



Electrode and electrolyte production

Cell assembly

Figure SC-BAT-SRB- 21: Aging (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

Aging is the last step in cell synthesis and is used to ensure quality. Changes in cell characteristics or performance are monitored throughout aging by measuring the open circuit voltage of the cell under controlled air conditions on a regular basis. The cells are kept in towers and/or aging shelves.

Additional information

Because the solid electrolyte allows the cell to reach stable characteristics more quickly than lithium-ion batteries, a shorter aging time is expected.

Process parameters & requirements

- State of charge of the cell at the beginning of aging
- Aging duration
- Controlled atmospheric conditions

Quality features [excerpt]

- Capacity
- Internal resistance
- Self-discharge
- Coulomb efficiency

Challenges [excerpt]

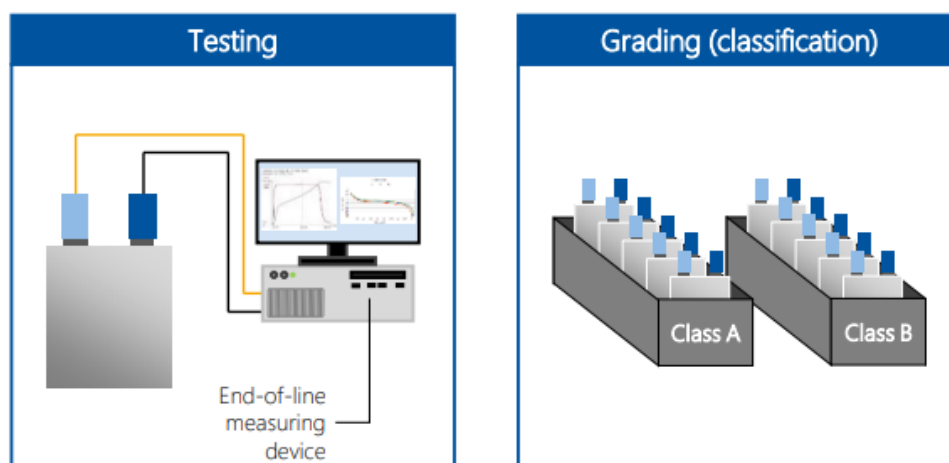
- Packing density of the cell workpiece carriers

Technology alternatives [excerpt]

- Aging is largely based on extensive experiential knowledge, which does not yet exist for all-solid-state batteries due to lack of series production maturity.

Transferability of lithium-ion battery cell manufacturing expertise

3.3.6 Cell Finishing: Testing and Grading



Electrode and electrolyte production

Cell assembly

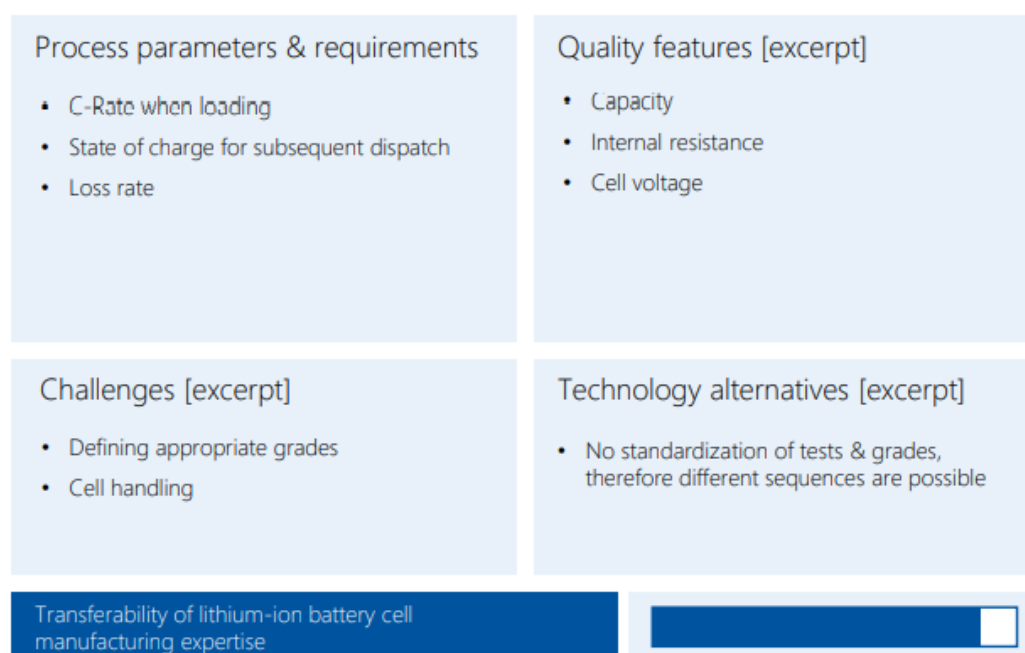
Figure SC-BAT-SRB- 22: Testing and Grading (Source: Masterdokument_Zellherstellung_v63 Vordruckfreigabe.pptx (rwth-aachen.de))

Production process

The attributes of the produced battery cell are checked in the last step. This comprises, above all, the characterization of the electrochemical properties, in addition to an optical check for damage. The all-solid-state batteries are graded according to their electrochemical attributes based on the measurement findings and the capacity established during creation.

Additional information

Because there is no grading system, different grades do not allow for direct judgments regarding quality, but rather aggregate cells with comparable characteristics.



Sample Topic Activities

Activity 1: Video Learning

Instruction to Teacher: ‘

Play the following video about building a solid-state battery for the students to get an idea of how they can do it themselves as projects or subject of interest in further studies.

Link: <https://youtu.be/N7yYlvQo5jQ>

Activity 2: Group Presentation

Instruction to Teachers

1. Group Students
2. Task Groups to research and select different solid-state batteries of interest. Their research must cover battery design, composition, working principle or mechanism, and application. Students should also justify how their solid-state battery of interest can revolutionize Battery Technology, Electronics, and Energy Industry.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 3: Practical (Optional)

Title: Graphite Circuit: An Electricity Science Experiment with just 3 items

This experiment is just a demonstrate to show the conductivity of solid graphite from a pencil being used to power a small LED Light bulb. From this basic knowledge, students should see the potential of designing a solid-state batter.

Materials

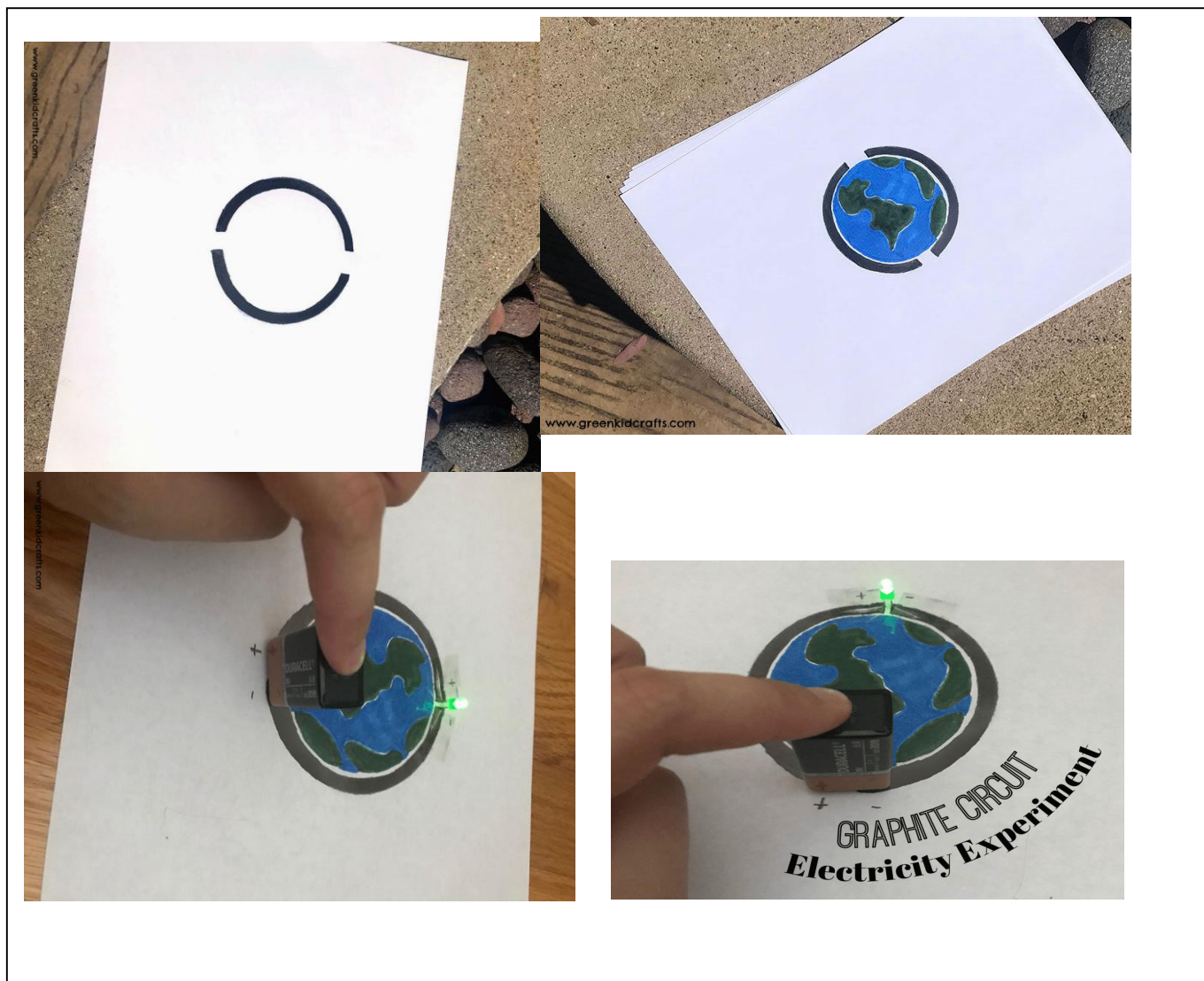
- 1 Graphite Art Pencil
- Paper
- Tape
- Mini LED Light Bulb
- 9V Battery

Instructions

1. On a piece of paper, sketch out your graphite circuit design. You may start by drawing a form and then add the graphite afterwards. Make care you draw a form with two apertures on either end. This will be vital in the construction of our graphite circuit!
2. Create a thick line of graphite over your shape. Add positive and negative symbols to the two open areas as guides. We colored the inside of the circle like the earth but you can draw other things too!
3. Tape the wires of your LED bulb to the graphite opening aligning the long wire with the positive side of the opening and the shorter wire with the negative side of the opening. Tape the bulb in an upright position. This is where those + and – symbols we drew earlier come in handy! Place your 9v battery on the opposite end over the positive and negative sides of the graphite. The light bulb should light up! Ta-da! We have completed our graphite circuit and created an electrical current. Pretty dang awesome huh?

How does this Electrical Graphite Circuit Work?

Electricity has the ability to flow from one place to another along a path. A circuit is a closed path from one place to another like a loop. We're creating an electrical circuit with the graphite, light bulb, and battery. The graphite acts as a path for the electrical energy. When the battery is placed on the graphite, energy flows from the battery, along the graphite path, through the wires on the light bulb, continuing back to the battery completing the circuit. If the battery is removed, the circuit is broken. This is also true if you remove the light bulb. We use the light bulb to show us the electricity that is flowing along the path.



References

(n.d.). Retrieved from http://samples.jbpub.com/9781449600914/86632_CH02_027_074.pdf

ashajyoti. (2021, August 30). *N terminus and c terminus in protein*. Retrieved from ScienceQuery.com: <https://sciencequery.com/n-terminus-and-c-terminus-in-protein/>

Parkinson, Giles. (2021, October 10). *Andrew Forrest to build massive hydrogen, wind, solar and cabling factory in Queensland*. Retrieved from reneweconomy.com.au: <https://www.google.com/amp/s/reneweconomy.com.au/forrest-to-build-massive-hydrogen-wind-solar-and-cabling-factory-in-queensland/amps/>

PNG Business News. (2021, August 11). *PNG SIGNS AGREEMENT WITH FORTESCUE METALS GROUP*. Retrieved from PNG Business News Website: www.pngbusinessnews.com/articles/2021/8/png-signs-agreement-with-fortescue-metals-group

Protein Sequencing. (2021, October 31). Retrieved from Wikipedia : https://en.wikipedia.org/wiki/Protein_sequencing#:~:text=Protein%20sequencing%20is%20the%20practical%20process%20of%20determining,identify%20the%20protein%20or%20characterize%20its%20post-translational%20modifications.

- [Masterdokument_Zellherstellung_v63_Vordruckfreigabe.pptx \(rwth-aachen.de\)](#)
- [Robot Programming: A Guide to Programming Your Own Robots \(edgy.app\)](#)
- [Nanorobotics \(fractal.org\)](#)
- [Making and Operating Molecular Machines: A Multidisciplinary Challenge - Baroncini - 2018 - ChemistryOpen - Wiley Online Library](#)
- [Driving Smart Molecular Systems by Artificial Molecular Machines \(wiley.com\)](#)
- [Masterdokument_Zellherstellung_v63_Vordruckfreigabe.pptx \(rwth-aachen.de\)](#)
- <https://www.scienceinschool.org/article/2016/chembot-chemistry-robots/>
- [Robot Programming: A Guide to Programming Your Own Robots \(edgy.app\)](#)
- [Robotics Designing of PCB - javatpoint](#)
- <https://blog.finology.in/Legal-news/Ways-to-Commercialize-IPR>
- <https://www.upcounsel.com/why-do-patents-expire>
- <https://courses.lumenlearning.com/wmopen-introductiontobusiness/topic/intellectual-property-2/>
- <https://www.forbes.com/sites/arielcohen/2021/02/11/what-batteries-will-power-the-future/?sh=5ce1b64841c0>
- <https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fcourses.mak.ac.ug%2Fsites%2Fdefault%2Ffiles%2Fdownloads%2FIndustrial%2520Biochemistry.doc&wdOrigin=BROWS ELINK>
- <https://www.biochemistry.org/education/careers/becoming-a-bioscientist/what-is-biochemistry/>
- <https://chemintelligence.com/ai-for-chemistry#:~:text=Artificial%20Intelligence%20%28AI%29%20is%20being%20used%20more%20and, and%20the%20time%20to%20market%20for%20new%20drugs.>
- <https://www.javatpoint.com/facebook-data-mining>
- <https://www.cs.princeton.edu/courses/archive/fall05/cos597A/lectures/fundamentals.pdf>
- [Introduction to Protein Electrophoresis | Bio-Rad Laboratories](#)
- http://samples.jbpub.com/9781449600914/86632_ch02_027_074.pdf
- [Microsoft Word - Protein Gel Electrophoresis.doc \(cpb-us-e1.wpmucdn.com\)](#)

- <https://ccsuniversity.ac.in/bridge-library/pdf/Fundamentals-Medicinal-Chemistry-2003-By-Gareth-Thomas.pdf>
- [A sweet out-of-the-box solution to the hydrogen economy: is the sugar-powered car science fiction? - Energy & Environmental Science \(RSC Publishing\)](#)
[DOI:10.1039/B818694D](#)
- [Grey, blue, green – the many colours of hydrogen explained | World Economic Forum \(weforum.org\)](#)
- [The Hydrogen Economy | Introduction to Chemistry \(lumenlearning.com\)](#)
- [Hydrogen Is the Fuel Of The Future. Or Is It? \(cnbctv18.com\)](#)
- [Fuel Cell - Definition, Working, Types, and Applications of fuel cell. \(byjus.com\)](#)
- Credit of this Project: Eleftheria Karagiorgau, Vasileio Spahos.
<https://www.scienceinschool.org/article/2016/chembot-chemistry-robots/>